Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students [1]. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people [2].

Introduction courses to fundamental mathematical topics usually look like this: here'e are some basic definitions, let's derive some specific results from them - now go apply these in your field.

[1] and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

[2] the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

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Why Write Yet Another Book About Linear Algebra?

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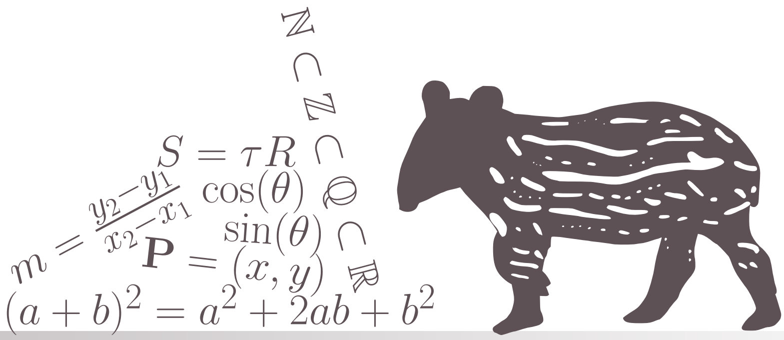
CHAPTER

1 MATHEMATICAL BACKGROUND



1.1 PREFACE

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Testing some equations: let $\vec{x}, \vec{y} \in \mathbb{R}^n$ and $\alpha, \beta \in \mathbb{R}$. Then from the fact that $\mathbb{R}^n$ is a vector space, we get

$$\vec{c} = \alpha\vec{x} + \beta\vec{y} \in \mathbb{R}^n. \tag{1.1}$$

However, if $\vec{a} \in \mathbb{R}^m$, where $m \neq n$, then $\alpha\vec{x} \in \mathbb{R}^m$, and therefore $\vec{c}$ is not defined (i.e. Equation 1.1 is irrelevant).

Now, this is a test of the *aligned* environment, which allows for multiple (correctly-aligned) lines of equations and having the label only once (instead of per line, as in the *align* environment):

$$\begin{aligned}
\sin(2x) &= \frac{1}{2\mathrm{i}}\left(\mathrm{e}^{2x\mathrm{i}} - \mathrm{e}^{-2x\mathrm{i}}\right) \\
&= \frac{1}{2\mathrm{i}}\left(\left[\mathrm{e}^{x\mathrm{i}}\right]^2 - \left[\mathrm{e}^{-x\mathrm{i}}\right]^2\right) \\
&= \underbrace{\frac{1}{2\mathrm{i}}\left(\mathrm{e}^{x\mathrm{i}} - \mathrm{e}^{x\mathrm{i}}\right)}_{\sin(x)}\underbrace{\left(\mathrm{e}^{x\mathrm{i}} + \mathrm{e}^{x\mathrm{i}}\right)}_{2\cos(x)} \\
&= 2\sin(x)\cos(x).
\end{aligned} \tag{1.2}$$

Note 1.1 Test note

This is a test note. I will try to also refernce it later.

!

Now, I wish to refernce Note 1.1. Did it work?

Also, let us look at an example.

Example 1.1 Some test example

If $x = \begin{bmatrix} 1 & 2 & -3 \end{bmatrix}$ and $y = \begin{bmatrix} 5 & -1 \end{bmatrix}$, then $n = 3$ and $m = 2$. Clearly, $n \neq m$ - and so $\alpha\vec{x} \in \mathbb{R}^3$, while $\beta\vec{y} \in \mathbb{R}^2$. This means that $\alpha\vec{x} + \beta\vec{y}$ is not defined.

...let's check if referencing an example works correctly: can you see Example 1.1?

I believe it is now time for some figures! See Figure 1.1.

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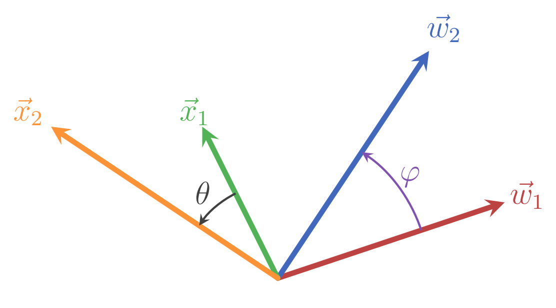

Figure 1.1: This is the first test figure. Does it look ok?

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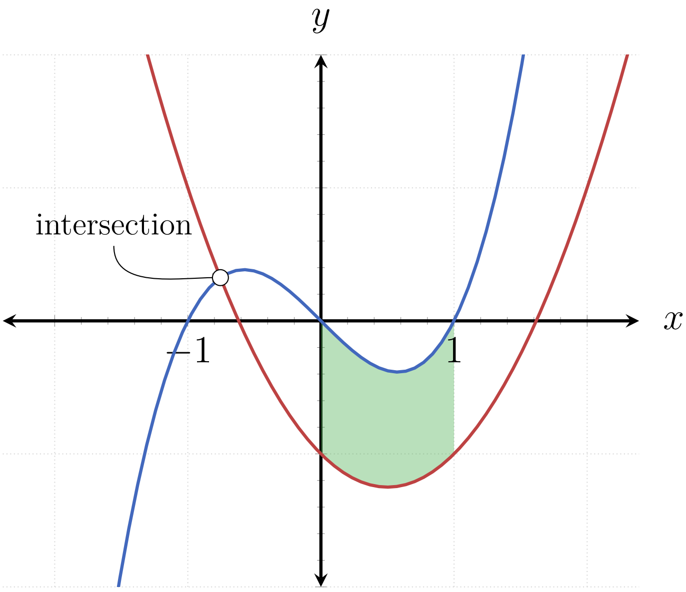

Figure 1.2: And this is another test figure! I hope it works.

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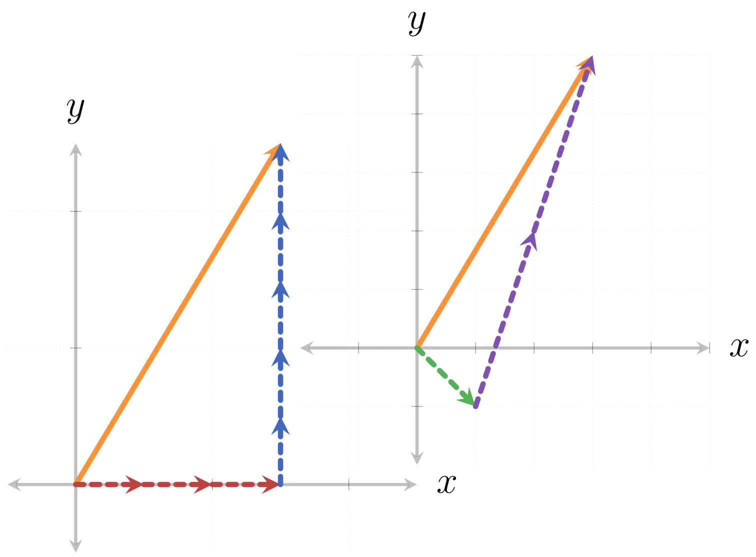

Figure 1.3

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$$T(\alpha\vec{x} + \beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} \mid \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

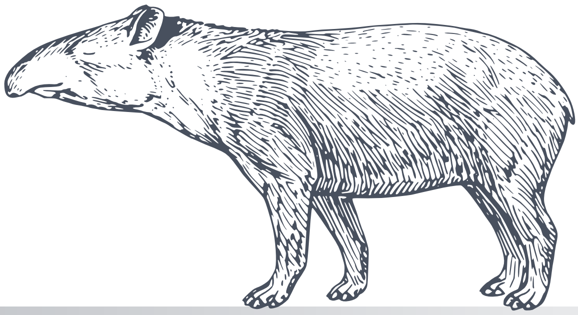
$$(\mathbf{A} \cdot \mathbf{B})^{\intercal} = \mathbf{B}^{\intercal} \cdot \mathbf{A}^{\intercal}$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A}\vec{x} = \lambda\vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c^i_j = \langle a^i \mid b_j \rangle$$



CHAPTER 2

THE BASICS OF LINEAR ALGEBRA

2.1 INTRODUCTION AND MOTIVATION

2.2 VECTORS AND VECTOR SPACES

2.2.1 MOTIVATION

Motivating Problem 2.1 Gravity Force

Consider a simple simulation of classical ("Newtionian") gravity, say of a planet with mass m orbiting a star with mass M (where $M \gg m$). At any given moment, the force acting on the planet due to the star's gravity has magnitude

$$F = G\frac{Mm}{r^2}, \tag{2.1}$$

where r is the distance between the planet and the star at that moment, and G is a universal constant (specifically, $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$).

However, since we live in 3-dimensional space[a], we also want this force to be pointing in the direction from the planet to the star. To do this, we need to understant how to describe orientations in 3-dimensional space, and be able to pack together both orientation and magnitude in the correct way.

[a] allegedly.

Motivating Problem 2.2 Bouncing/Reflecting Off a Wall

Imagine we are writing a 3D computer game in which some object is bouncing off a wall. Alternatively, consider 3D graphics in general, where one simulates rays of light reflecting off of smooth surfaces. In both cases the same problem needs to be solved: given a flat plane of some orientation in space, and an object hitting it from some direction - what would be the direction of the object after it bounces off the plane?

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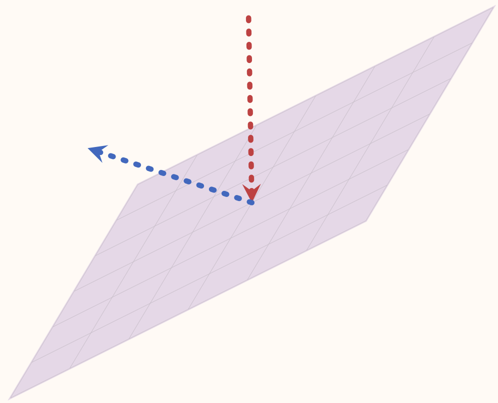

To solve this problem we will need to know how to represent quantities like position, velocity and orientation in 3D, and more generally points, lines and planes - and to know how to calculate the intersection between these different objects. By the end of this section you should be able to solve this problem by yourself, and even generalize it to any number of dimensions.

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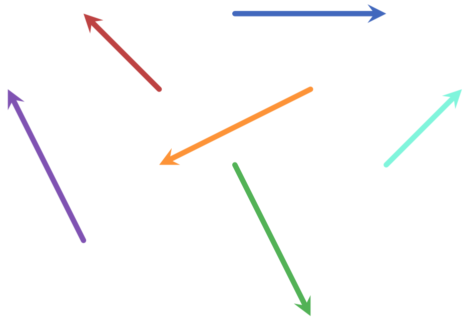
2.2.2 VECOTRS AND THEIR GEOMTERIC PROPERTIES

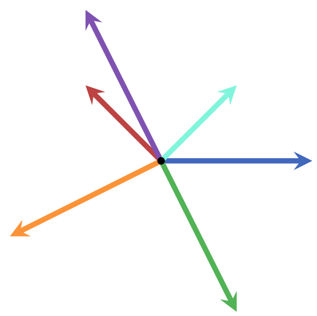
The above motivating problems require objects that have both a magnitude and a direction: for example, the force of gravity between a star and a planet has magnitude given by Equation 2.1 and a direction from the planet to the star. In Motivating-problem 2.2, if we model some object hitting a wall, the object's velocity is a combination of magnitude (which we usually simply call speed) and a direction (towards the wall).

In the simplest terms, vectors are mathematical objects which have both a magnitude and a direction. The magnitude of a vector is called norm, while its direction is sometimes refered to as its orientation. Vecotrs can exist in 2-dimesnsions, 3-dimensions or even higher dimensions. Since higher dimensions are rather hard to conseptulize, I will first focus on 2- and 3-dimensional vectors so that we can better undetstand their properties, and only then generalize to higher dimensions when it's relevant.

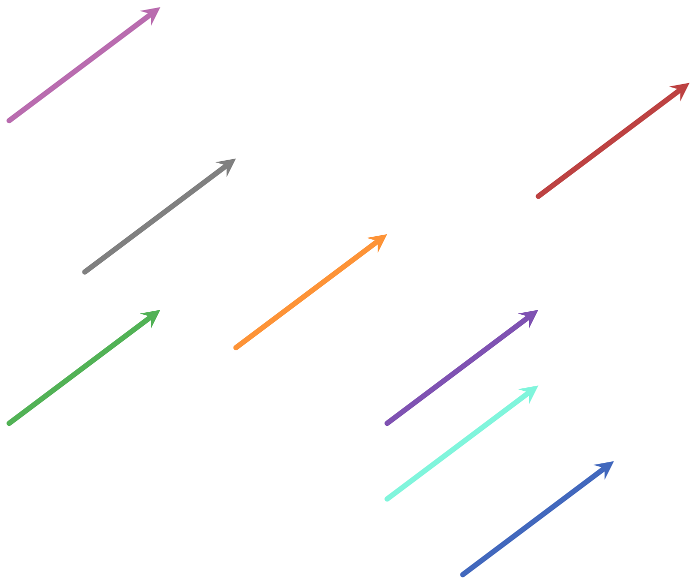
In 2-dimensions we can represent vectors as arrows:



However, since we are usually interested in the magnitude and direction of vectors and not where they originate from, we draw them as originating in the same location which we unimaginatively call the origin:



In fact, all vectors that have the same norm and orientation are considered identical - no matter where they originate. For example, the following vectors are actually all identical:



Since we will be using vectors as mathematical objects and would like to use them in an arithmetical context, we need to use some kind of notation. Usually, vectors are denoted as lower-case Latin letters, either in boldface or with a line or an arrow symbol above or below the letter. Here are few examples:

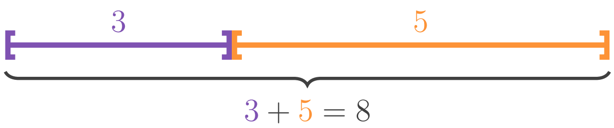
$$\begin{gathered}
\mathbf{a},\ \mathbf{b},\ \mathbf{c},\ \mathbf{x},\ \mathbf{y},\ \mathbf{z},\ \ldots \\
\overline{a},\ \overline{b},\ \overline{c},\ \overline{x},\ \overline{y},\ \overline{z},\ \ldots \\
\underline{a},\ \underline{b},\ \underline{c},\ \underline{x},\ \underline{y},\ \underline{z},\ \ldots \\
\vec{a},\ \vec{b},\ \vec{c},\ \vec{x},\ \vec{y},\ \vec{z},\ \ldots \\
\vec{\mathbf{a}},\ \vec{\mathbf{b}},\ \vec{\mathbf{c}},\ \vec{\mathbf{x}},\ \vec{\mathbf{y}},\ \vec{\mathbf{z}},\ \ldots
\end{gathered}$$

In this book I am using the last notation - a boldface Latin letter with an arrow above it. This ensures that it is both easy to read and is distinctive enough to quickly differentiate vectors from other objects. Sometimes, depending on context, the vectors are colored to make it easier to follow through (the process of setting up/developing a concept/equation).

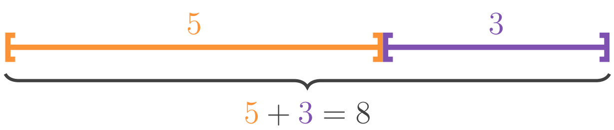
Just like with real numbers, we would like to be able to peform two basic opertaions with vectors: addition and multiplication. Since vectors are fundamentaly geometric objects, we can take inspiration from how we perform these two operations on real numbers geometrically.

- **Vector addition**: there are some properties that we would like vector addition to have: the simplest is that if we add a vector and its opposite, they should cancel each other out somehow. Another property is that averaging two vectors makes sense: ...

The addition of two real numbers can be interpreted geometrically as follows: each number is represented by a horizontal line segment of appropriate length: the number 2 is represented by a line segment of length 2, the number 5 is represented by a line segment of length 5, etc. Addition of two numbers is done by placing intervals side-by-side and measuring their total length. For example, the sum of 3 and 5 can be represented by placing side-by-side two intervals, of of lengths 3 and 5 respectively:



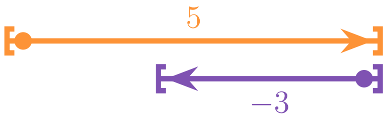
Note that the order of summation doesn't matter: $3 + 5 = 5 + 3 = 8$, and this is evident in the geometric method:



Subtraction also works pretty well using this method, but we have to tweak it just a little to make things more clear: we'll add a direction to our intervals - right for positive and left for negative. So for example, the number 4 is positive, and therefore as a "direction" to the right:

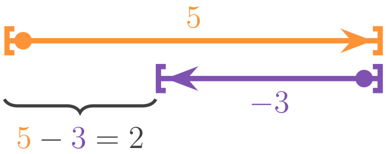
Its opposite -4 is represented with a line segment of the same length but directed to the left:

Now we can rewrite subtraction as addition with negative numbers. For example, $5 - 3 = 5 + (-3)$:



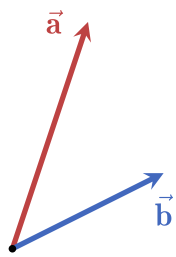
(I'm drawing the line segments here with a vertical offset so that we can visually see what's going on - instead of the segments covering each other)

The result of the summation works exactly the same as in the simple positives-only case: it's the distance from the start of 5 to the end (head) of 3:



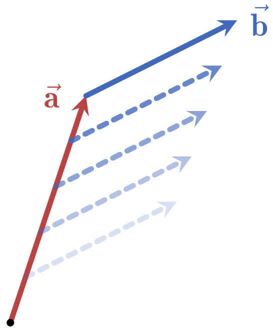
...

any two vectors of the same dimensionality (e.g. 2-dimensional, 3-dimensional, 4-dimensional, etc.) can be added together, and the result is a third vector. Pictorially, here are two vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ which we want to sum together:

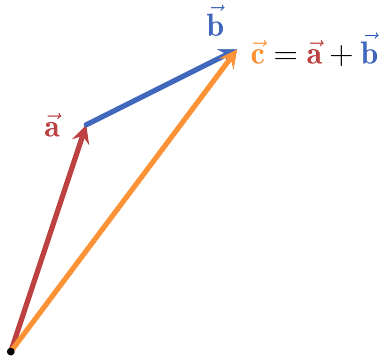


The steps to produce the sum of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is the following:

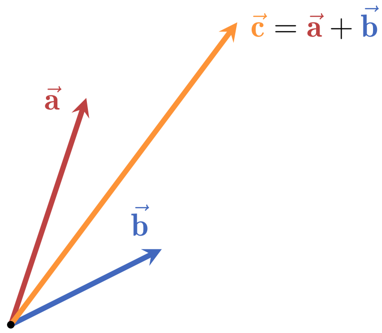
1. move $\vec{\mathbf{b}}$ such that its origin aligns with the head of $\vec{\mathbf{a}}$:



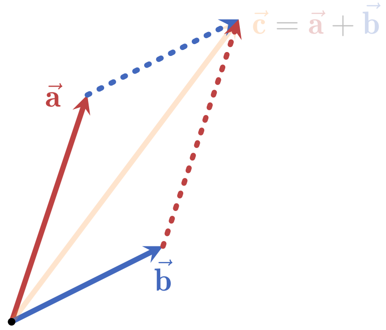
2. draw a new vector from the origin to the head of the shifted $\vec{\mathbf{b}}$:



3. the resulting vector $\vec{\mathbf{c}}$ *is* the sum of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$. Here are all three vectors together from the same origin:

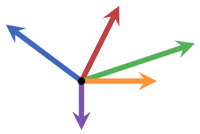


This method is called the triangle method, and it displays an important property of vector addition: it is commutative! This means that the order of addition doesn't matter: $\vec{\mathbf{a}} + \vec{\mathbf{b}} = \vec{\mathbf{b}} + \vec{\mathbf{a}}$. To see this visually, we can apply the triangle method twice, i.e. moving both $\vec{\mathbf{a}}$ *and* $\vec{\mathbf{b}}$:

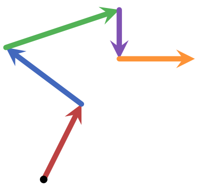


> We see that no matter which vector we add onto which (i.e. what is the order of addition) - the result is the same. The shape formed by this demonstration is called a parallelogram, and it is incredibly important in linear algebra as a whole as we will see in future sections.

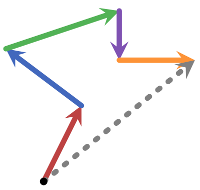
This method of summing vectors can be applied to any number of vectors. For example, consider the following 5 vectors:



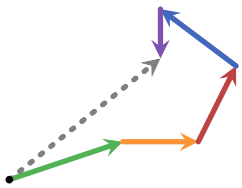
To sum them together, all we need is to start with one the vectors (say, the red one) and use the triangle method to add the other vectors step by step:



Finally, we connect the origin of the first vector with the head of the last vector (in this case the orange one):

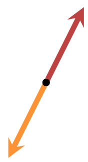


and that's it, the dashed gray vector is the sum of all five vectors. To see that the order of addition really doesn't matter we can do this again but this time starting with the green vector and changing the summation order completely:



As can be seen, the resulting dashed gray vector is exactly the same as before.

Another important property of the addition of vectors is that we can always take a vector and rotate it by 180°, giving us a vector that "cancels" the original vector by summation. For example, the following red vector is rotated to yield its opposite, colored in orange:



These vectors added together using the triangle method yield a vector with 0 norm:

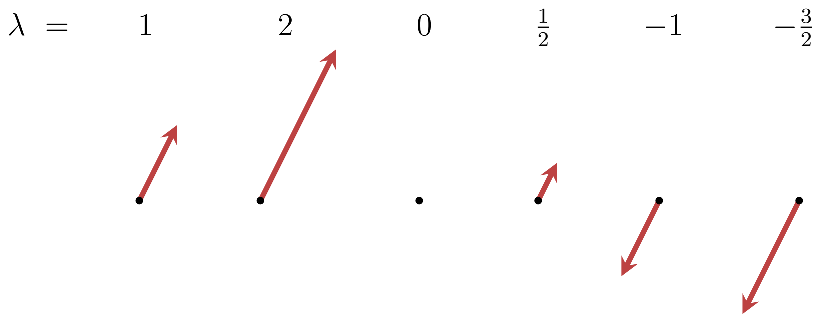
(here I drew the vectors a bit shifted from each other so that we can see both)

Such a vector is called the zero vector, denoted as $\vec{\mathbf{0}}$. It has no direction, and a norm of 0. Adding $\vec{\mathbf{0}}$ to any other vector is the same as not adding any vector at all (and it doesn't matter "from which side" we add $\vec{\mathbf{0}}$):

$$\vec{\mathbf{a}} + \vec{\mathbf{0}} = \vec{\mathbf{0}} + \vec{\mathbf{a}} = \vec{\mathbf{a}}. \tag{2.2}$$

We say that the zero vector is *agnostic* (also: neutral) to addition.

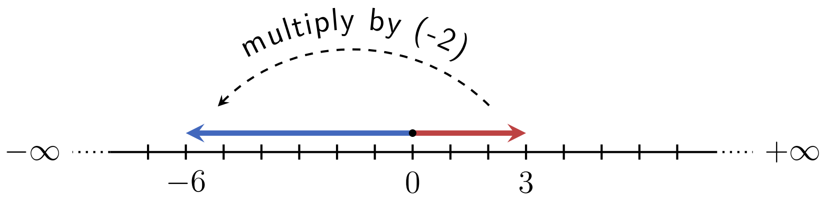
- **Scaling vectors**: any vector can be scaled by any real number, which in this context we call a scalar. Scaling by a number simply "stretches" the vector by the number. For example, in the following figure a red vector is scaled by a scalar λ, taking on different values:



This example shows us several things:

- no matter by which positive scalar we scale a vector, it always points in the same direction.
- If we scale a vector by 0 it becomes the zero vector.
- Finally, scaling a vector by a negative scalar rotates the vector by 180° and then scales it by the absolute value of the scalar.

This is similar to how real numbers behave under multiplication: multiplying a real number by a negative number is, in a sense, "flipping" and scaling it relative to 0 (the origin). For example, multiplying 3 by -2 flips the the number to point to the left, and then scales it to be of length 6:



Therefore, scaling a vector is denoted just like multiplication of real numbers: $\lambda\vec{\mathbf{a}}$, where λ is the scalar. In the above example, if the red vector is $\vec{\mathbf{a}}$, then the other five vectors can be written as $2\vec{\mathbf{a}}$, $0\vec{\mathbf{a}}$, $\frac{1}{2}\vec{\mathbf{a}}$, $-1\vec{\mathbf{a}}$ and $-\frac{3}{2}\vec{\mathbf{a}}$.

Furthermore, we can place the vector notation in different places and even hiding the scalar depending on convinience, so for example we can write $-\vec{\mathbf{a}}$ instead of $-1\vec{\mathbf{a}}$ and $-\frac{3\vec{\mathbf{a}}}{2}$ instead of $-\frac{3}{2}\vec{\mathbf{a}}$.

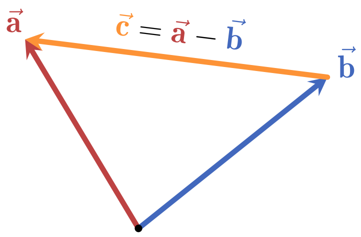
Just like with real numbers, we call $-\vec{\mathbf{a}}$ the additive inverse of $\vec{\mathbf{a}}$.

In fact, the similarities to real number don't end there: subtracting a vector $\vec{\mathbf{b}}$

from another vector $\vec{\mathbf{a}}$ is simply adding $\vec{\mathbf{a}}$ and the additive inverse of $\vec{\mathbf{b}}$:

$$\vec{\mathbf{a}} - \vec{\mathbf{b}} = \vec{\mathbf{a}} + (-\vec{\mathbf{b}}). \tag{2.3}$$

Visually, if we place the origin of $\vec{\mathbf{a}} - \vec{\mathbf{b}}$ at the head of $\vec{\mathbf{b}}$, its head would be at the same point as the head of $\vec{\mathbf{a}}$:



While subtration is not commutative, it is antisymmetric:

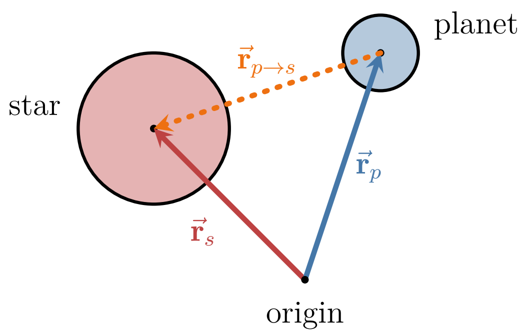
$$\vec{\mathbf{a}} - \vec{\mathbf{b}} = -\left(-\vec{\mathbf{a}} + \vec{\mathbf{b}}\right) = -\left(\vec{\mathbf{b}} + (-\vec{\mathbf{a}})\right) = -\left(\vec{\mathbf{b}} - \vec{\mathbf{a}}\right), \tag{2.4}$$

i.e. flipping the order of subtraction changes the sign of the expression.

So we can use vector subtraction as a convinient way to point from one vector to another. This can be utilized to implement the gravity force in Motivating-problem 2.1: if we know the position of the planet (let's call it $\vec{\mathbf{r}}_p$) and the position of the star ($\vec{\mathbf{r}}_s$), a vector pointing from the planet's center to the star's center would be, for example,

$$\vec{\mathbf{r}}_{p\to s} = \vec{\mathbf{r}}_p - \vec{\mathbf{r}}_s. \tag{2.5}$$

(the notation $p \to s$ is just an easy way to represent that the vector is pointing from the planet to the star)



However, this vector has a norm equal to the distance between the plant and the star - and we want to use it to calculate the *force* acting on the planet, which has a different magnitude (and units). To do this, we need to be able to scale $\vec{\mathbf{r}}_{p\to s}$ to whatever norm we want. Taking inspiration from real numbers, we can employ the following procedure: say we want to scale some non-zero number x to be y. We can multiple x by $\frac{y}{x}$, canceling the x:

$$x \mapsto \cancel{x} \cdot \frac{y}{\cancel{x}} = y.$$

A different way to phrase this is that we simply multiplying (or *scaling*) x by $\frac{1}{x}$, and then multiply the result by y:

$$x \mapsto \cancel{x} \cdot \frac{1}{\cancel{x}} \cdot y = y.$$

We can apply the same technique to vectors. Scaling a vector by its *reciprocal norm* yields a vector with norm 1, which we denote by replacing the arrow in the vector notation by a "hat":

$$\hat{\mathbf{a}} = \frac{1}{\|\vec{\mathbf{a}}\|}\vec{\mathbf{a}}. \tag{2.6}$$

This operation is called normalization, and the resulting vector is a unit vector (also: normalized vector).

Now, if we scale a unit vector $\hat{\mathbf{a}}$ by any scalar λ we get a vector of the same direction as $\hat{\mathbf{a}}$ (and also $\vec{\mathbf{a}}$), and with a norm of λ. The vectors $\vec{\mathbf{a}}$, $\hat{\mathbf{a}}$ and $\lambda\hat{\mathbf{a}}$ are all parallel to each other.

Back to our star-planet system: by normalizing the vector $\vec{\mathbf{r}}_{p\to s}$ we get a unit vector pointing from the planet to the star: $\hat{\mathbf{r}}_{p\to s}$, and scaling by the magnitude of the gravitational attraction (Equation 2.1) we finally get the correct *gravity force vector*:

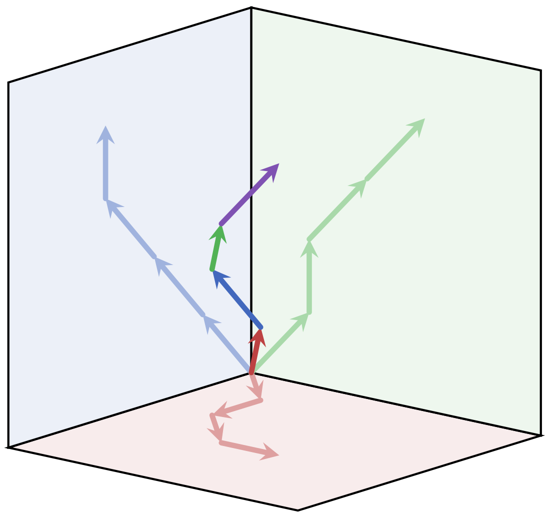
$$\underbrace{\boxed{\vec{\mathbf{F}}}}_{\text{vector}} = \underbrace{\boxed{G\frac{Mm}{r^2}}}_{\text{scalar}} \underbrace{\boxed{\hat{\mathbf{r}}_{p\to s}}}_{\text{vector}} \tag{2.7}$$

Indeed, $\vec{\mathbf{F}}$ is a vector: it has a magnitude (the scalar $G\frac{Mm}{r^2}$) and a direction (the unit vector $\hat{r}_{p\to s}$). Any vector can be written as a unit vector scaled by a real number:

$$\vec{\mathbf{a}} = a\hat{\mathbf{a}}, \tag{2.8}$$

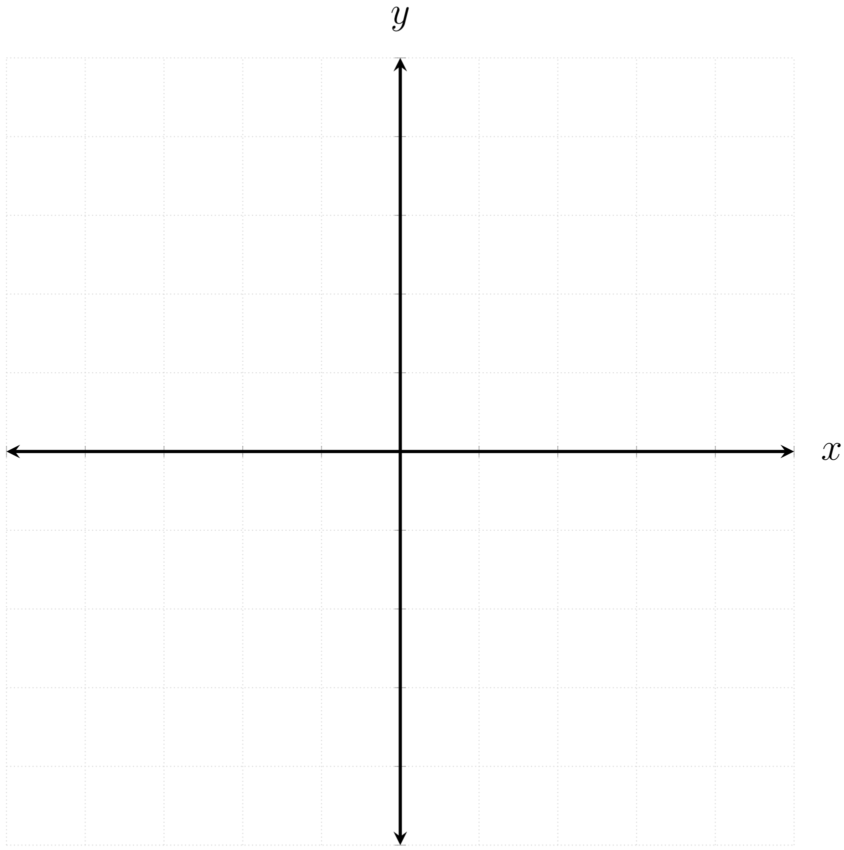
where $a \in \mathbb{R}$ is the magnitude of $\vec{\mathbf{a}}$, i.e. $a = \|\vec{\mathbf{a}}\|$. This is nothing more than a rearrangement of Equation 2.6: by dividing both sides of the equation by $a = \|\vec{\mathbf{a}}\|$ we get the definition of a unit vector.

TBW: text about how in 3D the same rules apply.



2.2.3 CARTESIAN COORDINATE SYSTEM AND BASIS SETS

Everything we discussed so far is very nice - but it doesn't give us much in the way of actually doing any practical calculations. To do that, we need to "quantify" vectors somehow. By far the most common way in 2- and 3-dimensions is to use a Cartesian coordinate system: in dimensions we simply choose a direction to be the x-direction (usually the horizontal direction such that the numbers increase to the right), and then the choose between the two possible perpendicular directions to be the positive y-direction:



We then set the origin of all vectors to be at the origin of the axes (i.e. the point $(0, 0)$). To quantify a vector, we draw two lines from its head: one parallel to the x-axis and one parallel to the y-axis. The points at which these lines intersect the y- and x-axes are then, respectively, the x- and y-coordinates of the vector: for example, Figure 2.1 shows a vector $\vec{\mathbf{a}}$ placed in a Cartesian coordinate system. The x-coordinate of the vector is $a_x = 2$ and its y-coordinate is $a_y = 3$.

Using these coordinates, we can then write $\vec{\mathbf{a}} = (3, 2)$, i.e. identify the vector with the point in the Cartesian system where its head is (given that its origin aligns with the origin of the system). In general, this works for any vector: each point $\mathbf{P} = (x, y)$ in the Cartesian system can be identified with a single vector $\vec{\mathbf{v}} = (x, y)$, and each vector can be identified with a single point.

However, vector are not points, and we should make this fact reflected in the notation we use. Therefore I choose to use the column-form notation for vectors,

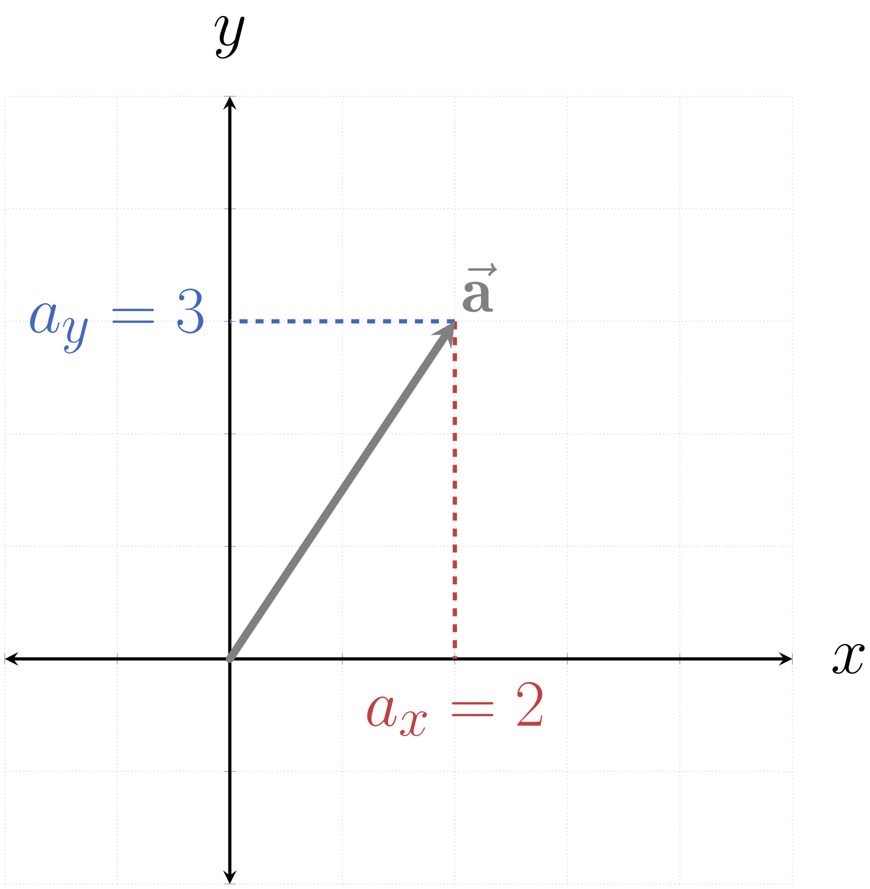


Figure 2.1: The vector $\vec{\mathbf{a}}$ placed in a Cartesian coordinate system, making its coordinates $a_x = 3$ and $a_y = 2$.

for example

$$\vec{\mathbf{a}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \ \vec{\mathbf{v}} = \begin{bmatrix} x \\ y \end{bmatrix}, \tag{2.9}$$

etc. In principle, I could have chosen to use a similar notation, namely the row-form notation:

$$\vec{\mathbf{a}} = \begin{bmatrix} 2 & 3 \end{bmatrix}, \ \vec{\mathbf{v}} = \begin{bmatrix} x & y \end{bmatrix}. \tag{2.10}$$

While both notations are essentially equivalent, to keep notational consistency I would need to strictly use one of these notations, and for reasons of personal preferences I chose the column-form. This enforces the use of some notational conventions later, for example the meaning of matrix columns vs. rows and the notation for a set of related objects called covectors (discussed in details in REF).

Let's see how the two basic properties of vectors (magnitude and direction) look like using the column notation. Figure 2.2 shows a vector $\vec{\mathbf{v}}$ in a Cartesian coordinates system. The triangle $\triangle\mathbf{OVA}$ is a right triangle, with the side $\overline{\mathbf{OV}}$ being equal to the x-coordinate of $\vec{\mathbf{v}}$ and the side $\overline{\mathbf{VX}}$ equal to the y-coordinate of $\vec{\mathbf{v}}$. We can calculate the length of the hypotenuse $\overline{\mathbf{OV}}$ using the Pythagorean theorem:

$$\overline{\mathbf{OV}}^2 = \overline{\mathbf{OX}}^2 + \overline{\mathbf{VX}}^2. \tag{2.11}$$

Replacing each line with the respective vector component, and then taking the square root we get

$$\|\vec{\mathbf{v}}\| = \sqrt{v_x^2 + v_y^2}. \tag{2.12}$$

How does the column form of a vector behaves under the two basic vector operations (addition and scaling)? Let's see what happenes when we add two vectors together, say

$$\vec{\mathrm{a}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \vec{\mathrm{b}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Following the triangle method for vector addition, we first $\vec{\mathrm{b}}$ such that its origin lies at the head of $\vec{\mathrm{a}}$, and the resulting addition, $\vec{\mathrm{c}}$, is drawn from the origin to the head of $\vec{\mathrm{b}}$. In Figure 2.3 we can see the projection of both $\vec{\mathrm{a}}$, $\vec{\mathrm{b}}$ and $\vec{\mathrm{c}}$ onto the x- and y-axes, respectively: each component of $\vec{\mathrm{c}}$ is simply the addition of the same

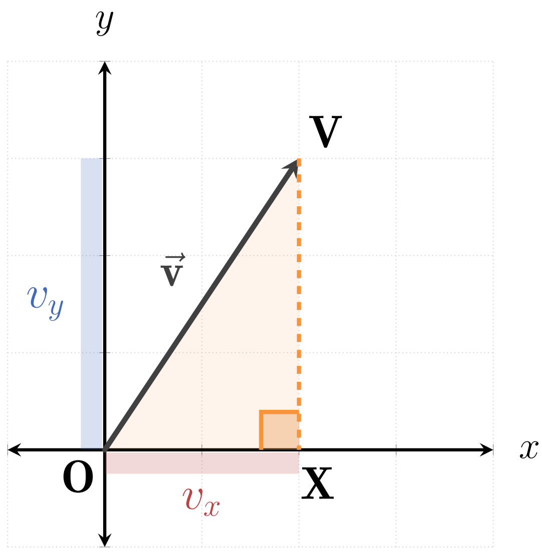


Figure 2.2: A vector $\vec{\mathbf{v}}$ with components v_x and v_y creates a right triangle between the origin $\mathbf{O}$, the point $\mathbf{X} = (v_x, 0)$ and the point $\mathbf{V} = (v_x, v_y)$. Therefore the norm of the vector can be derived using the Pythagorean theorem.

components of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$:

$$\vec{\mathbf{c}} = \vec{\mathbf{a}} + \vec{\mathbf{b}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+2 \\ 3+1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}. \tag{2.13}$$

In general, additing two vectors together …

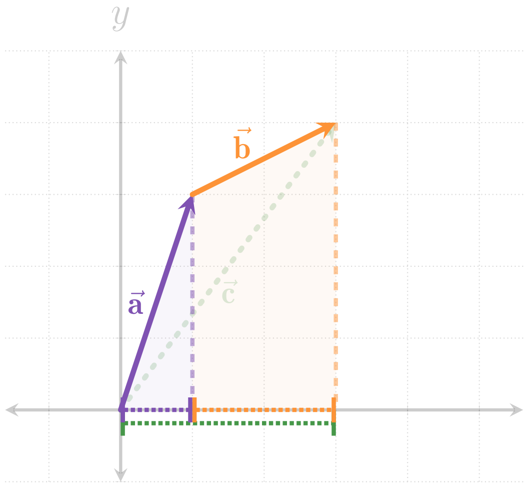
$$\begin{bmatrix} a_x \\ a_y \end{bmatrix} + \begin{bmatrix} b_x \\ b_y \end{bmatrix} = \tag{2.14}$$

This result is also known as component-wise addition.

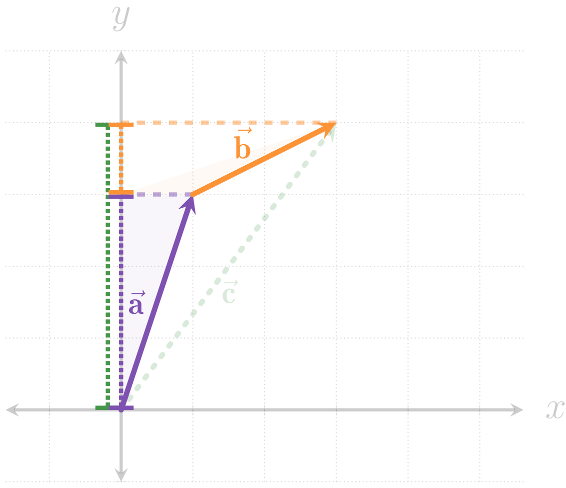
…

Another way of interpreting the coordinates of a vector is to define two unit vectors:

$$\hat{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{2.15}$$



(a) Projection on x-axis.



(b) Projection on y-axis.

Figure 2.3: Projecting three vectors on the x- and y-axes: $\vec{\mathbf{a}}$, $\vec{\mathbf{b}}$ and $\vec{\mathbf{c}} = \vec{\mathbf{a}} + \vec{\mathbf{b}}$. Note how the projection of $\vec{\mathbf{c}}$ on to an axis is exactly the same as the addition of the projections of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ on the same axis.

Looking at Figure 2.4, these two vectors correspond to the two axes of the Cartesian coordinate system: $\hat{x}$ is a unit vector in the direction of the x-axis, and $\hat{y}$ is a unit vector in the direction of the y-axis.

2.2.4 GEOMTERIC OBJECTS

2.2.5 PROJECTIONS AN‘D REJECTIONS

2.2.6 VECTOR SPACES

2.2.7 SOLVING THE MOTIVATING PROBLEMS

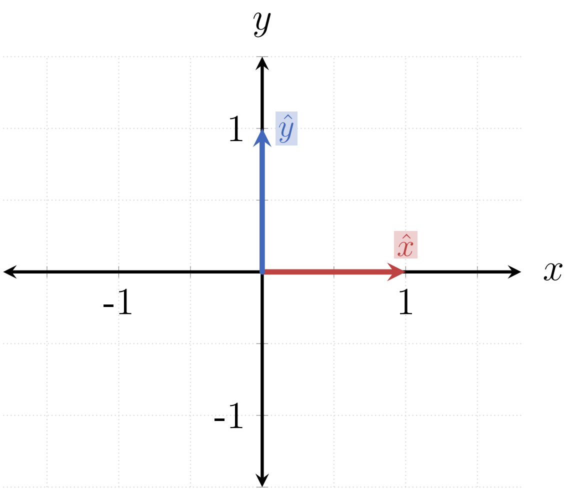


Figure 2.4

2.3 LINEAR TRANSFORMATIONS

Summary 2.1 The Next Two Sections

This section and the one immediately following it deal with a rather simple, yet very powerful type of functions between vector spaces: linear transformations.
In this section we will:

1. see why are we interested in linear transformations,
2. what are their properties, and
3. how they can provide a powerful tool for many different applications.

In the next section we will explore how we can turn the caulculations of linear transformations into an extremely easy and straight-forward process using matrices.

2.3.1 INTRODUCTION

Text text text...

...

The main purely mathematical property of linear transformations is that they preserve the structure of vector spaces. This means that if we have three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ such that $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, then for any linear transformation T,

$$T(\vec{c}) = \alpha T(\vec{a}) + \beta T(\vec{b}). \tag{2.16}$$

(see Figure 2.5 for a visual example)

However, this is a rather abstract way of discussing linear transformations. Instead, let's first learn some of the more "graphical" properies of linear transformations in the plane, i.e. $T : \mathbb{R}^2 \to \mathbb{R}^2$, and then see practical examples of such transformations before moving on to their more abstract (and powerful) mathematical description.

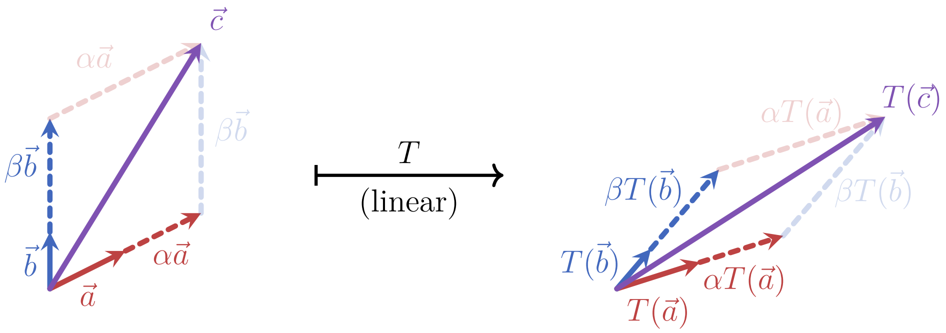


Figure 2.5: Two vectors $\vec{a}$ and $\vec{b}$ are both scaled (by α and β, respectively) and then added together to form the vector $\vec{c}$. All the addition and scaling relations between the vectors are kept under the application of a linear transformation T.

2.3.2 LINEAR TRANSFORMATIONS IN 2-DIMENSIONS

(here comes an introductory text about the fundamental linear transformations in $\mathbb{R}^2$, what are their properties and some more relevant information...)

Any linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ always has the following four properties:

- **The origin is kept untouched**: T always maps $\vec{0}$ to $\vec{0}$.

- **Lines remain lines**: a set of points forming a line in $\mathbb{R}^2$ will be mapped to another set of points forming a line (with possibly different direction, distance from origin and scale).

- **Parallel directions remain parallel**: if $\vec{a}$ and $\vec{b}$ point at the same direction, then $\vec{b} = \alpha\vec{a}$ for some $\alpha \in \mathbb{R}$. After the transformation the maps of $\vec{a}$ and $\vec{b}$ will stay parallels, and would even keep their scalar relation: $T(\vec{b}) = \alpha T(\vec{a})$.

- **All areas are scaled by the same factor**: if two shapes S_1 and S_2 have areas A_1 and A_2, respectively, before the transformation is applied, their maps $\tilde{S}_1$ and $\tilde{S}_2$ would have the areas kA_1 and kA_2 (where $k \in \mathbb{R}$ depends on the transformation and can take both positive, zero and negative values).

Figure 2.6 shows these properties in action: there are two planes in the figure - before and after the application of a random linear transformation T (up and down, respectively). The following can be seen:

1. The origin stays at the same place - see how a small purple dot placed on the origin doesn't change its position after the application of T.

2. All lines (e.g. axes, grid lines, the two purple lines) are mapped to lines in the transformed plane.

3. Parallel directions remain parallel, e.g. grid lines aligned with one of the axes, and the two purple lines.

4. All areas are scaled by the same factor: notice for example how all unit squares formed by the grid lines (one is highlighted in orange) are mapped to identical parallelograms. You can also check to see that the blue rectangle is exactly twice as large than the orange square before and after the transformation.

The example in Figure 2.6 also shows what *doesn't* get preserved under a linear transformation. To see this more clearly, Figure 2.7 "zooms-in" on the orange unit square: we label its vertices as $\mathbf{O}, \mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ starting from the point at the origin and going anti-clockwise. The transformed vertices are then, respectively, $\mathbf{O}', \mathbf{A}', \mathbf{B}'$ and $\mathbf{C}'$.

In the original unit square the diagonals are equal: their lengths are $\mathbf{OB} = \mathbf{AC} = \sqrt{2}$. However, it's clear that $\mathbf{O'B'} < \mathbf{A'C'}$. In addition, all the angles in the original unit square are, of course, right angles, but the transformed square is a parallelogram ($\mathbf{O'A'B'C'}$) has two equal acute angles of value α and two equal obtuse angles of value β (where $\beta = \pi - \alpha$).

So we can see that at least two quantities aren't necessarily conserved by linear transformations: (1) **distances**, which unlike areas don't even all scale by the same amount, and (2) **angles** (more generally, **directions**).

One last key observation before we move on: as can be seen in Figure 2.6 and Figure 2.7 linear transformations in $\mathbb{R}^2$ transform **squares** into **parallelograms**. These parallelograms can be scaled, "squeezed", reflected and/or rotated compared to the original squares.

There are several "basic" linear transformations in $\mathbb{R}^2$: these transformations can be thought of as the building blocks of all possible linear transformations on a 2-dimensional plane. We will see later how we can combine these basic transformations to createnew transformations, but first let's introduce these "basic" linear transformations.

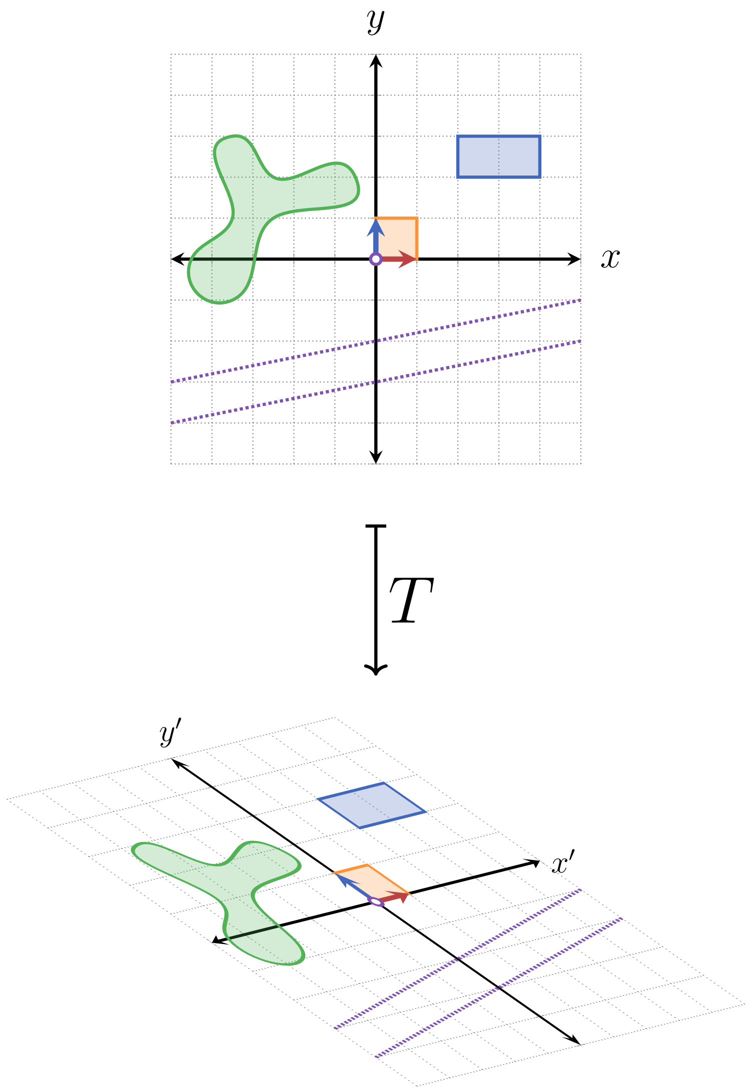


Figure 2.6: Some random linear transformation T applied to $\mathbb{R}^2$, with different shapes showing the four main properties discussed for linear transformations.

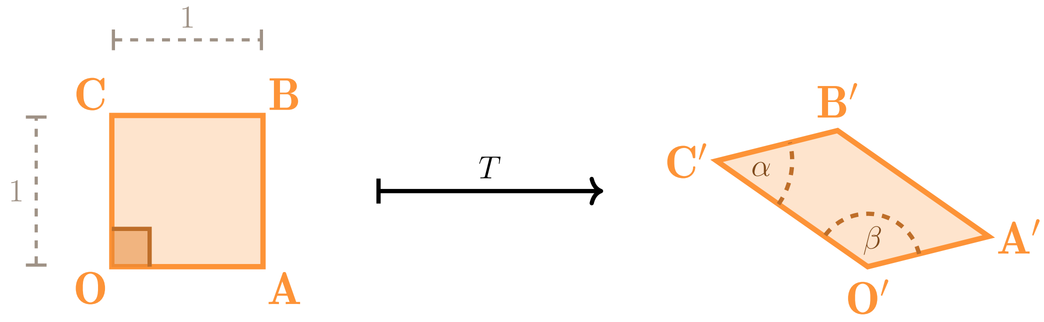


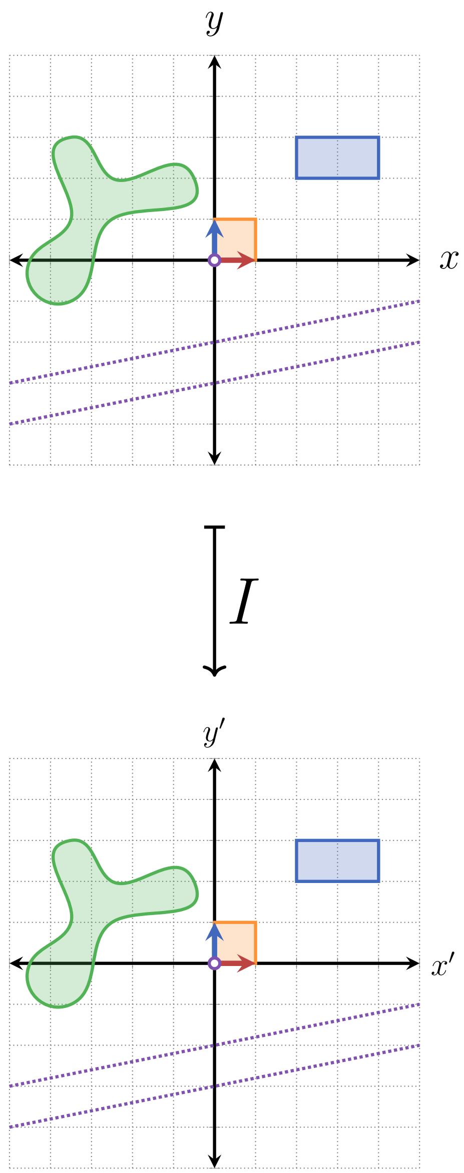
Figure 2.7: Unit square in $\mathbb{R}^2$ before and after the application of the random linear transformation T from Figure 2.6.

In the following introduction, each transformation is shown using a drawing of $\mathbb{R}^2$ with several shapes on it: the two standard basis vectors $\hat{x}$ and $\hat{y}$, a unit area between the two vectors (drawn in orange), a blue rectangle, two purple parallel lines, a somewhat random green shape[1] and the grid lines (or rather, grid dots). When looking at these figures you should pay attention to how each shape is deformed, and especially what relationship are kept the same or are changed: for example, parallel lines remain parallel - but other angles between lines (including 90°) are not kept. Also note the areas of the different shapes and how they change under the transformation.

Let's begin:

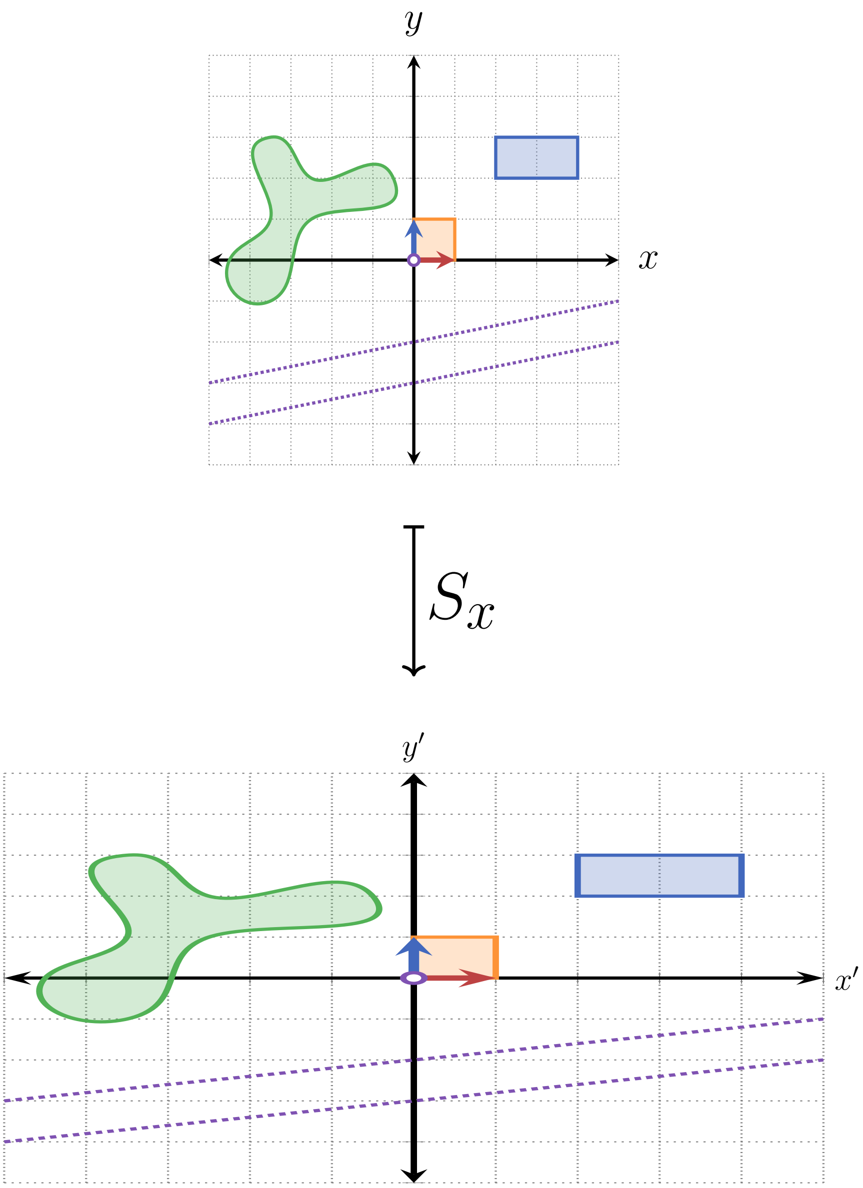
- **The identity transformation**: not doing anything to the plane is a transformation, even if a rather boring one:

[1] I shamelessly stole the shape from https://tikz.org/examples/chapter-12-drawing-smooth-curves/ because it's perfect for this illustration.



Indeed, the identity transformation has all of three key properties of linear transformations: the origin is perserved, parallel lines remain parallel after the transformation, and all areas are scaled by the same factor (namely 1).

- **Scaling**: Scaling the plane can be done in either the x-diretion or the y-direction. Just as with vectors, the scaling factor can be any real number, positive or negative (where negative scaling factors flip the direction of the axis):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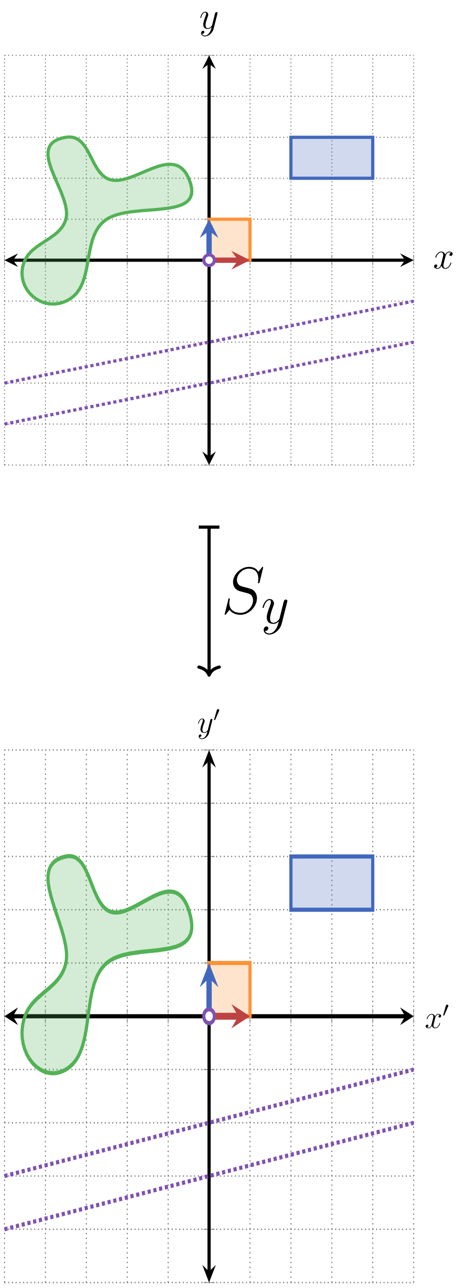
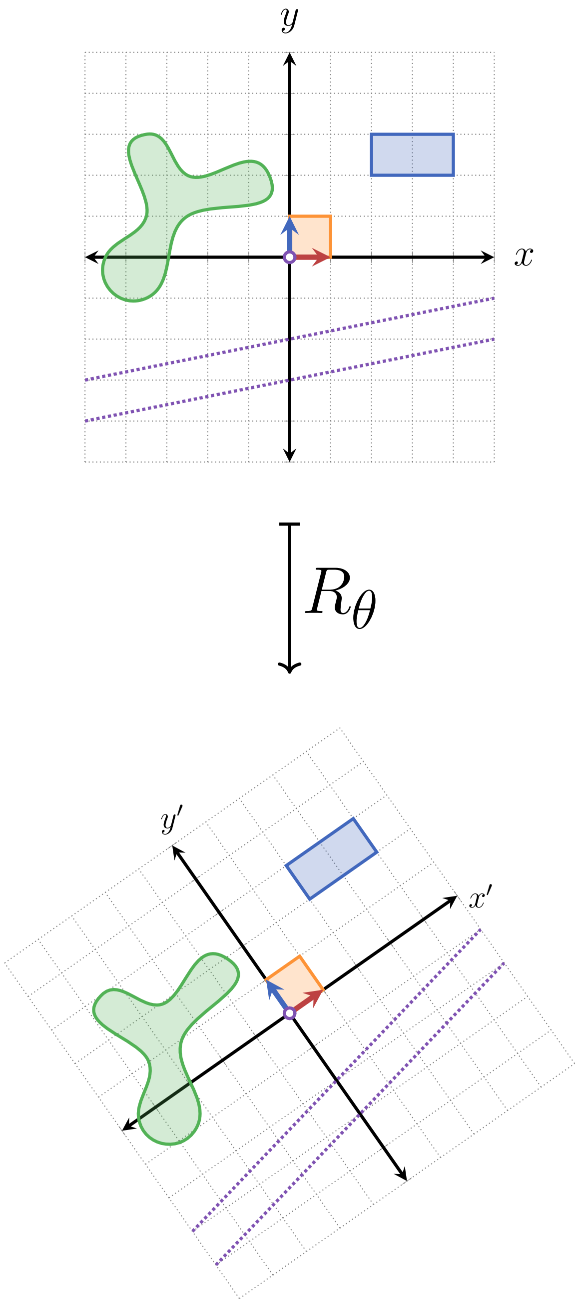
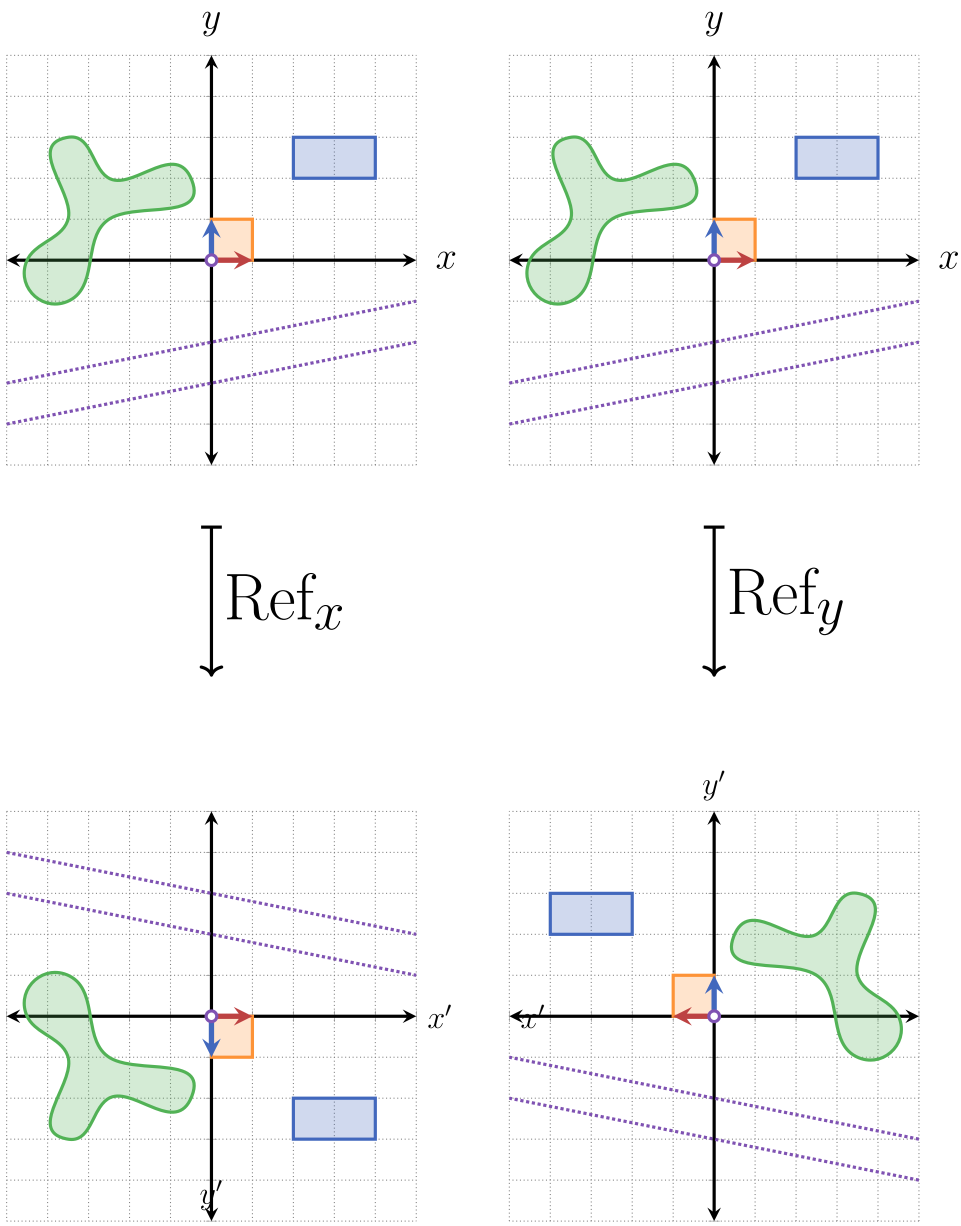


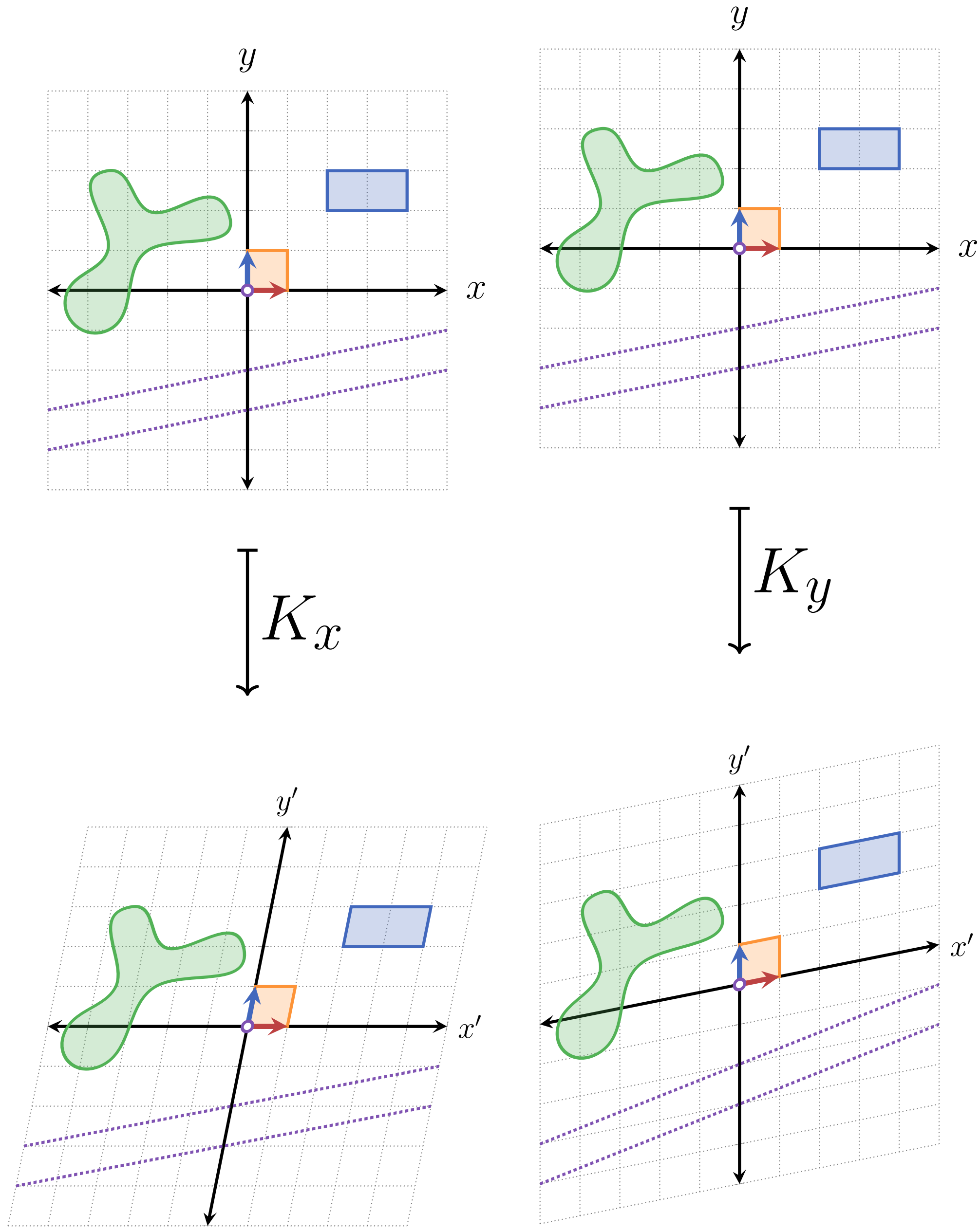
- **Rotation about the origin, anti-clockwise**: text text text:



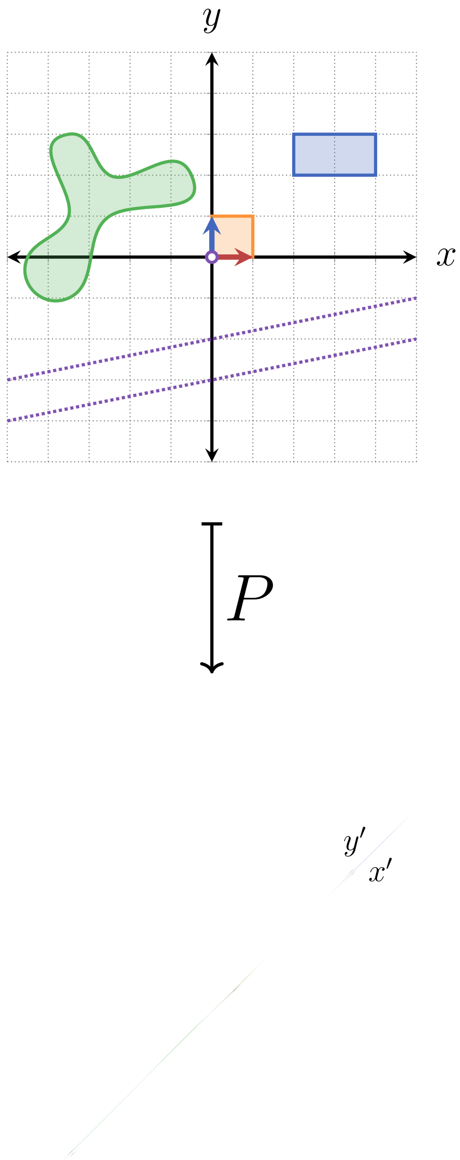
- **Reflection across a line through the origin**: text text text:



- **Shear**: text text text:



- **Projection on line through the origin**: text text text: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2.3.3 EXAMPLES OF NON-LINEAR TRANSFORMATIONS IN 2-DIMENSIONS

2.3.4 3-DIMENSIONAL LINEAR TRANSFORMATIONS

2.3.5 FORMAL DEFINITION

The formal definition of linear transformations between any two vector spaces is the following:

Definition 2.1 Linear Transformation

A transformation $T:\mathbb{R}^n\to\mathbb{R}^m$ is said to be *linear* if for any two vectors $\vec{a},\vec{b}\in\mathbb{R}^n$ and any scalar $\alpha\in\mathbb{R}$:

1. $T\left(\vec{a}+\vec{b}\right)=T(\vec{a})+T(\vec{b})$, and
2. $T\left(\alpha\vec{a}\right)=\alpha T(\vec{a})$.

We can combine the above two criteria using an additional scalar $\beta\in\mathbb{R}$: T is linear if

$$T(\alpha\vec{a}+\beta\vec{b})=\alpha T(\vec{a})+\beta T(\vec{b}).$$

π

Just like with vectors, the first condition is called **additivity** and the second condition is called **homogeneity**. What these conditions say, in simple words, is that we can always scale vectors and add them together before we apply a linear transformation - and the result will be the same as first transforming each vector separately, and then scaling and/or add them together. So in practice we can choose what we want to calculate first (addition and scaling or the transformation) according to what is easier and/or more convinient to calculate.

The formal definition of linear transformations can be used to prove the first three of the four basic geometric properties of linear transformations in 2- and 3-dimensions. In fact, this works for any n-dimensional Euclidean space $\mathbb{R}^n$, so let's prove the more general form of these properties instead of limiting ourselves to 2- or 3-dimensions.

Proof 2.1 Geometric Properties of Linear Transformations

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. Then:

- **The origin is kept untouched**: the origin in $\mathbb{R}^n$ is represented by the n-dimensional zero vector, $\vec{0}$. We can write the zero vector as $0\vec{v}$, where $\vec{v}$ is any vector in $\mathbb{R}^n$ (this is just scaling a vector by 0, which gives the zero vector). When we apply T to $0\vec{v}$ we get

$$T(0\vec{v}) = 0T(\vec{v}) = \vec{0},$$

since $T(\vec{v})$ is a vector in $\mathbb{R}^n$, so scaling it by 0 gives $\vec{0}$.

- **Lines remain lines**: A line in $\mathbb{R}^n$ can be represented just like in 2- and 3-dimensions (see REF TO LINES IN PREVIOUS SECTION):

$$\mathbf{L}(s) = \vec{x}_0 + s\hat{d}.$$

Applying T to $\mathbf{L}$ gives us

$$T\left(\mathbf{L}(s)\right) = T\left(\vec{x}_0 + s\hat{d}\right),$$

and by the linearity of T we can separate the right side of the equality:

$$T\left(\vec{x}_0 + s\hat{d}\right) = T(\vec{x}_0) + sT(\hat{d}).$$

Since both $T(\vec{x}_0)$ and $sT(\hat{d})$ are vectors in $\mathbb{R}^n$, the result of adding them is also a vector in $\mathbb{R}^n$:

$$T(\vec{x}_0) + sT(\hat{d}) = \vec{a} + s\vec{b} = \mathbf{C}(s),$$

where $\mathbf{C}(s)$ is now a function of s with a line structure (vector + variable times a direction) - proving that T transforms lines into lines.

- **Parallel directions remain parallel**: parallel lines have the same direction, i.e. if $\mathbf{L}_1(s)$ and $\mathbf{L}_2(s)$ are parallel, then they can be written as

$$\begin{aligned}\mathbf{L}_1(s) &= \vec{x}_1 + s\hat{d},\\ \mathbf{L}_2(s) &= \vec{x}_2 + s\hat{d}.\end{aligned}$$

(here $\vec{x}_1$ and $\vec{x}_2$ might be different, but $\hat{d}$ - the direction of the lines - is the same for both)

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Applying the linear transformation T on these lines gives

$$T\left(\mathbf{L}_1\right) = T\left(\vec{x}_1 + s\hat{d}\right) = T\left(\vec{x}_1\right) + sT\left(\hat{d}\right),$$
$$T\left(\mathbf{L}_2\right) = T\left(\vec{x}_2 + s\hat{d}\right) = T\left(\vec{x}_2\right) + sT\left(\hat{d}\right),$$

and whatever the value of $T(\hat{d})$ is - it is the same direction for both transformed lines. We therefore get that T transforms two parallel lines (i.e. with the same direction) to two parallel lines.

QED

Unfortunately, the last property is rather difficult to prove with the material covered so far, so this proof will have to wait until a later part of the book (REF?).

Let's see how to use the formal definition of linear transformations to actually prove that a given transformation is indeed linear. We can start with the identity transformation since it's a very simple transformation.

- **Additivity**: for any $\vec{a}, \vec{b} \in \mathbb{R}^2$, we have

$$I(\vec{a}) = \vec{a},\ \ I(\vec{b}) = \vec{b}, \tag{2.17}$$

and also that

$$I\left(\vec{a} + \vec{b}\right) = \vec{a} + \vec{b}. \tag{2.18}$$

Therefore, if we first add the vectors $\vec{a}$ and $\vec{b}$ together and then apply I on the result, we get

$$I\left(\vec{a} + \vec{b}\right) \overset{\text{from Equation 2.18}}{=} \vec{a} + \vec{b} \underset{\text{from Equation 2.17}}{=} I(\vec{a}) + I(\vec{b}). \tag{2.19}$$

- **Homogeneity**: for any vector $\vec{a} \in \mathbb{R}^2$ and scalar $\alpha \in \mathbb{R}$ we have

$$I\left(\alpha\vec{a}\right) = \alpha\vec{a}, \tag{2.20}$$

and together with Equation 2.17 (specifically its first equality) we get

$$I\left(\alpha\vec{a}\right) = \alpha\vec{a} = \alpha I(\vec{a}). \tag{2.21}$$

Fullfilling these two conditions proves that I is indeed a linear transformation.

Next, let's see that rotation (about the origin) is also a linear transformation, starting with the simple $90°$ anti-clockwise rotation. Recall that a 2-dimensional vector $\vec{a} = \begin{bmatrix} a_x \\ a_y \end{bmatrix}$ can be rotated by $90°$ anti-clockwise by exchanging its components and then adding a minus sign to the first component (REF TO TRICK):

$$\begin{bmatrix} a_x \\ a_y \end{bmatrix} \xmapsto[\text{(anti-clockwise)}]{90° \text{ rotation}} \begin{bmatrix} -a_y \\ a_x \end{bmatrix}.$$

Using this fact we can easily show that a $90°$ rotation is a linear transformation:

- **Additivity**: we need two vectors, let's call them

$$\vec{a} = \begin{bmatrix} a_x \\ a_y \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} b_x \\ b_y \end{bmatrix}.$$

Then, adding the vectors first and then rotating the result we get

$$\mathrm{Rot}_{90°}\left(\vec{a}+\vec{b}\right) = \mathrm{Rot}_{90°}\left(\begin{bmatrix} a_x \\ a_y \end{bmatrix} + \begin{bmatrix} b_x \\ b_y \end{bmatrix}\right) = \mathrm{Rot}_{90°}\left(\begin{bmatrix} a_x + b_x \\ a_y + b_y \end{bmatrix}\right) = \begin{bmatrix} -\left(a_y + b_y\right) \\ a_x + b_x \end{bmatrix}.$$

And if we first rotate the vectors and then add the result we get

$$\mathrm{Rot}_{90°}\left(\vec{a}\right) + \mathrm{Rot}_{90°}\left(\vec{b}\right) = \begin{bmatrix} -a_y \\ a_x \end{bmatrix} + \begin{bmatrix} -b_y \\ b_x \end{bmatrix} = \begin{bmatrix} -(a_y + b_y) \\ a_x + b_x \end{bmatrix},$$

exactly the same as before.

- **Homogeneity**: if we scale $\vec{a}$ by some scalar $\lambda \in \mathbb{R}$ first and then rotating the result we get

$$\mathrm{Rot}_{90°}\left(\lambda\vec{a}\right) = \mathrm{Rot}_{90°}\left(\lambda\begin{bmatrix} -a_y \\ a_x \end{bmatrix}\right) = \mathrm{Rot}_{90°}\left(\begin{bmatrix} \lambda a_x \\ \lambda a_y \end{bmatrix}\right) = \begin{bmatrix} -\lambda a_y \\ \lambda a_x \end{bmatrix}.$$

And if we first rotate $\vec{a}$ and then scale it by λ we get

$$\lambda\mathrm{Rot}_{90°}\left(\vec{a}\right) = \lambda\begin{bmatrix} -a_y \\ a_x \end{bmatrix} = \begin{bmatrix} -\lambda a_y \\ \lambda a_x \end{bmatrix},$$

again - the same result as before.

Together this means that rotating around the origin anti-clockwise by 90° is indeed a linear transformation.

Moving on, let's show that a rotation around the origin by *any* angle θ is a linear transformation. To start, we need to know how to transform a general 2-dimensional vector $\vec{a} = \begin{bmatrix} a_x \\ a_y \end{bmatrix}$, which we will find, of course, by using basic trigonometry (use Figure 2.8 to follow these steps). First, we calculate the norm of $\vec{a}$ (which we'll call $\mathbf{a}$ for convenience):

$$\mathbf{a} = \|\vec{a}\| = \sqrt{a_x^2 + a_y^2}. \tag{2.22}$$

We can then express $\vec{a}$ in polar coordinates (REF), as it's a more natural way of dealing with angles (and thus - rotations):

$$a_x = \mathbf{a}\cos(\alpha), \tag{2.23}$$
$$a_y = \mathbf{a}\sin(\alpha). \tag{2.24}$$

Rotating the $\vec{a}$ by θ yields a new vector $\vec{b}$ which as the same norm as $\vec{a}$ (since rotating a vector doesn't change its norm), and an angle $\beta = \alpha + \theta$. Therefore in polar coordinates $\vec{b}$ is

$$b_x = \mathbf{a}\cos(\beta) = \mathbf{a}\cos(\alpha + \theta), \tag{2.25}$$
$$b_y = \mathbf{a}\sin(\beta) = \mathbf{a}\sin(\alpha + \theta). \tag{2.26}$$

Using the trigonomitric identities for the *cos* and *sin* of a sum of two angles (REF), we get:

$$b_x = \mathbf{a}\cos(\alpha + \theta) = \mathbf{a}\left[\cos(\alpha)\cos(\theta) - \sin(\alpha)\sin(\theta)\right], \tag{2.27}$$
$$b_y = \mathbf{a}\sin(\alpha + \theta) = \mathbf{a}\left[\sin(\alpha)\cos(\theta) + \sin(\alpha)\cos(\theta)\right]. \tag{2.28}$$
$$\tag{2.29}$$

Now, we can express α in terms of the inverse trigonometric functions arccos and arcsin:

$$\alpha = \arccos\left(\frac{a_x}{\mathbf{a}}\right) = \arcsin\left(\frac{a_y}{\mathbf{a}}\right). \tag{2.30}$$

Substituting Equation 2.30 into Equation 2.29 we get

$$\begin{aligned}
b_x &= \mathbf{a}\left[\cos(\alpha)\cos(\theta) - \sin(\alpha)\sin(\theta)\right] \\
&= \mathbf{a}\left[\cos\left(\arccos\left(\frac{a_x}{\mathbf{a}}\right)\right)\cos(\theta) - \sin\left(\arcsin\left(\frac{a_y}{\mathbf{a}}\right)\right)\sin(\theta)\right] \\
&= \cancel{\mathbf{a}}\left[\frac{a_x}{\cancel{\mathbf{a}}}\cos(\theta) - \frac{a_y}{\cancel{\mathbf{a}}}\sin(\theta)\right] \\
&= a_x\cos(\theta) - a_y\sin(\theta),
\end{aligned} \tag{2.31}$$

$$\begin{aligned}
b_y &= \mathbf{a}\left[\sin(\alpha)\cos(\theta) - \sin(\alpha)\cos(\theta)\right] \\
&= \mathbf{a}\left[\sin\left(\arcsin\left(\frac{a_y}{\mathbf{a}}\right)\right)\cos(\theta) - \sin\left(\arcsin\left(\frac{a_y}{\mathbf{a}}\right)\right)\cos(\theta)\right] \\
&= \cancel{\mathbf{a}}\left[\frac{a_y}{\cancel{\mathbf{a}}}\cos(\theta) - \frac{a_x}{\cancel{\mathbf{a}}}\sin(\theta)\right] \\
&= a_x\sin(\theta) + a_y\cos(\theta).
\end{aligned} \tag{2.32}$$

To sum up this result, here it is again - this time in "mapping" notation:

$$\begin{bmatrix} a_x \\ a_y \end{bmatrix} \xmapsto{\mathrm{R}_\theta} \begin{bmatrix} a_x\cos(\theta) - a_y\sin(\theta) \\ a_x\sin(\theta) + a_y\cos(\theta) \end{bmatrix}. \tag{2.33}$$

Now that we finally have a formula for rotating any vector in $\mathbb{R}^2$ by any angle about the origin, we can finally prove that this is indeed a linear transformation.

- **Additivity**: let

$$\vec{a} = \begin{bmatrix} a_x \\ a_y \end{bmatrix},\ \vec{b} = \begin{bmatrix} b_x \\ b_y \end{bmatrix}.$$

Adding the two together gives

$$\vec{a} + \vec{b} = \begin{bmatrix} a_x + b_x \\ a_y + b_y \end{bmatrix},$$

and then rotating the result by θ yields

$$\mathrm{Rot}_\theta\left(\vec{a} + \vec{b}\right) = \begin{bmatrix} (a_x + b_x)\cos(\theta) - \left(a_y + b_y\right)\sin(\theta) \\ (a_x + b_x)\sin(\theta) + \left(a_y + b_y\right)\cos(\theta) \end{bmatrix}.$$

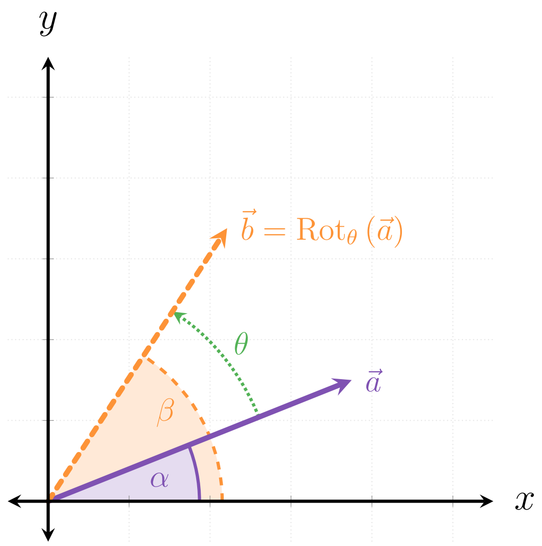

Figure 2.8: The vector $\vec{a}$ is rotated by an angle θ anti-clockwise about the origin, yielding the vector $\vec{b}$. $\vec{a}$ is at θ relative to the x-axis, which becomes $\beta = \alpha + \theta$ after the rotation.

On the other hand, rotating the two vectors first gives

$$\mathrm{Rot}_\theta\left(\vec{a}\right) = \begin{bmatrix} a_x \cos(\theta) - a_y \sin(\theta) \\ a_x \sin(\theta) + a_y \cos(\theta) \end{bmatrix},$$
$$\mathrm{Rot}_\theta\left(\vec{b}\right) = \begin{bmatrix} b_x \cos(\theta) - b_y \sin(\theta) \\ b_x \sin(\theta) + b_y \cos(\theta) \end{bmatrix}.$$

Finally, adding the two rotated vectors together we get

$$\begin{aligned} \mathrm{Rot}_\theta\left(\vec{a}\right) + \mathrm{Rot}_\theta\left(\vec{b}\right) &= \begin{bmatrix} a_x \cos(\theta) - a_y \sin(\theta) \\ a_x \sin(\theta) + a_y \cos(\theta) \end{bmatrix} + \begin{bmatrix} b_x \cos(\theta) - b_y \sin(\theta) \\ b_x \sin(\theta) + b_y \cos(\theta) \end{bmatrix}, \\ &= \begin{bmatrix} (a_x + b_x) \cos(\theta) - \left(a_y + b_y\right) \sin(\theta) \\ (a_x + b_x) \sin(\theta) + \left(a_y + b_y\right) \cos(\theta) \end{bmatrix}. \end{aligned}$$

- **Homogeneity**: Scaling $\vec{a}$ by a scalar $s \in \mathbb{R}$ and then rotating it by θ gives

$$\mathrm{Rot}_\theta\left(s\vec{a}\right) = \mathrm{R}_\theta \begin{bmatrix} sa_x \\ sa_y \end{bmatrix} = \begin{bmatrix} sa_x \cos(\theta) - sa_y \sin(\theta) \\ sa_x \sin(\theta) + sa_y \cos(\theta) \end{bmatrix}.$$

On the other hand, first rotating $\vec{a}$ and the scaling by s gives

$$s\mathrm{Rot}_{\theta}\left(\vec{a}\right) = s\begin{bmatrix} a_x\cos(\theta) - a_y\sin(\theta) \\ a_x\sin(\theta) + a_y\cos(\theta) \end{bmatrix} = \begin{bmatrix} sa_x\cos(\theta) - sa_y\sin(\theta) \\ sa_x\sin(\theta) + sa_y\cos(\theta) \end{bmatrix}.$$

Hoorah! Rotation is indeed a linear transformation.

2.3.6 COMPOSING LINEAR TRANSFORMATIONS

In the discussion about 2- and 3-dimensional linear transformations I mentioned several times compositions of such transformations and simply stated that they are also linear transformations. This topic, however, deserves some more exploration. First, in order for us to even be able to compose two transformations T_1 and T_2, the composition needs to be properly defined - specifically, if we want to compose T_2 onto T_1 (that is, first apply T_1 and then apply T_2 on the result), the **domain** of T_1 must be at least a superset of the **image** of T_1. Otherwise, we might get a situation where we apply T_1 to some vector and then not being able to apply T_2 on the result, since it isn't defined.

TBW: (1) more text about the domain and image of functions in relation to composition, (2) discussion on how the order of composition matters (i.e. it isn't commutative).

...

Proof 2.2 The Composition of Linear Transformations is Linear

Let $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^k$ be linear transformations, let $\vec{v}, \vec{w} \in \mathbb{R}^n$ and let $\lambda \in \mathbb{R}$. Then:

- **Additivity**:

$$\begin{aligned} \left(T_1 \circ T_2\right)\left(\vec{v} + \vec{w}\right) &= T_2\left(T_1\left(\vec{v} + \vec{w}\right)\right) \\ &= T_2\left(T_1(\vec{v}) + T_1(\vec{w})\right) \\ &= T_2\left(T_1(\vec{v})\right) + T_2\left(T_1(\vec{w})\right) \\ &= \left(T_1 \circ T_2\right)(\vec{v}) + \left(T_1 \circ T_2\right)(\vec{w}). \end{aligned}$$

- **Homogeneity**:

$$\left(T_1 \circ T_2\right)\left(\lambda\vec{v}\right) = T_2\left(T_1\left(\lambda\vec{v}\right)\right)$$

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$$\begin{aligned} &= T_2\left(\lambda T_1\left(\vec{v}\right)\right) \\ &= \lambda\Big(T_2\left(T_1(\vec{v})\right)\Big) \\ &= \lambda\left(T_1 \circ T_2\right)\left(\vec{v}\right). \end{aligned}$$

Therefore, $(T_1 \circ T_2)$ is also a linear transformation.

QED

2.3.7 GENERAL FORMS OF LINEAR TRANSFORMATIONS

You might have noticed by now that all of the linear transformations we dealt with had a similar structure, involving adding and scaling the different components of the vectors they act on. Indeed, the most general form of a linear transform in $\mathbb{R}^2$ is the following:

$$T\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix}, \tag{2.34}$$

where a, b, c and d are some real numbers. In a sense, this is just two linear combinations of the vector's components: one for the x-component and one for the y-component. Performing any other operation would make the transformation no longer linear. To see this, let's first quickly prove that this general form is indeed a linear transformation:

Proof 2.3 General 2-Dimensional Linear Transformation

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ a transformation defined for $\vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix} \in \mathbb{R}^2$ as

$$T\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix},$$

with $a, b, c, d \in \mathbb{R}$. As usual, we will prove that T confirms to the two criteria of linear transformations.

- **Additivity**: Let $\vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} w_x \\ w_y \end{bmatrix}$. Then first adding the two

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vectors together and then applying T we get

$$T(\vec{v}+\vec{w}) = T\begin{bmatrix} v_x + w_x \\ v_y + w_y \end{bmatrix} = \begin{bmatrix} a(v_x+w_x)+b(v_y+w_y) \\ c(v_x+w_x)+d(v_y+w_y) \end{bmatrix}.$$

And on the other hand, transforming each vector separately first and then adding the results, we get

$$\begin{aligned} T(\vec{v})+T(\vec{w}) &= \begin{bmatrix} av_x+bv_y \\ cv_x+dv_y \end{bmatrix} + \begin{bmatrix} aw_x+bw_y \\ cw_x+dw_y \end{bmatrix} \\ &= \begin{bmatrix} av_x+bv_y+aw_x+bw_y \\ cv_x+dv_y+cw_x+dw_y \end{bmatrix} \\ &= \begin{bmatrix} a(v_x+w_x)+b(v_y+w_y) \\ c(v_x+w_x)+d(v_y+w_y) \end{bmatrix}. \end{aligned}$$

- **Homogeneity**: Let $\vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ and $\lambda \in \mathbb{R}$. Then scaling $\vec{v}$ by λ and then applying T gives us

$$T(\lambda\vec{v}) = T\begin{bmatrix} \lambda v_x \\ \lambda v_y \end{bmatrix} = \begin{bmatrix} a\lambda v_x + b\lambda v_y \\ c\lambda v_x + d\lambda v_y \end{bmatrix} = \lambda\begin{bmatrix} av_x+bv_y \\ cv_x+dv_y \end{bmatrix}.$$

And applying T to $\vec{v}$ first and then scaling it by λ gives

$$\lambda T(\vec{v}) = \lambda T\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \lambda\begin{bmatrix} av_x+bv_y \\ cv_x+dv_y \end{bmatrix}.$$

Therefore, T is a linear transformation.

QED

2.3.8 CALCULATING LINEAR TRANSFORMATIONS, PART 1

Linear transformations have a very nice property which makes applying them on different vectors an extremely efficient process. Consider the linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ which is a composition of scaling by a factor of 2 in the y-diretion followed by a rotation of 90° anti-clockwise around the origin. Now let's say that we wish to calculate how T transforms the vector

$$\vec{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

In principle, we could simply apply the operations one after the other on $\vec{a}$:

$$\vec{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \xmapsto{\times 2 \text{ in } y} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \xmapsto{90^\circ \text{ rotation}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = T(\vec{a}). \tag{2.35}$$

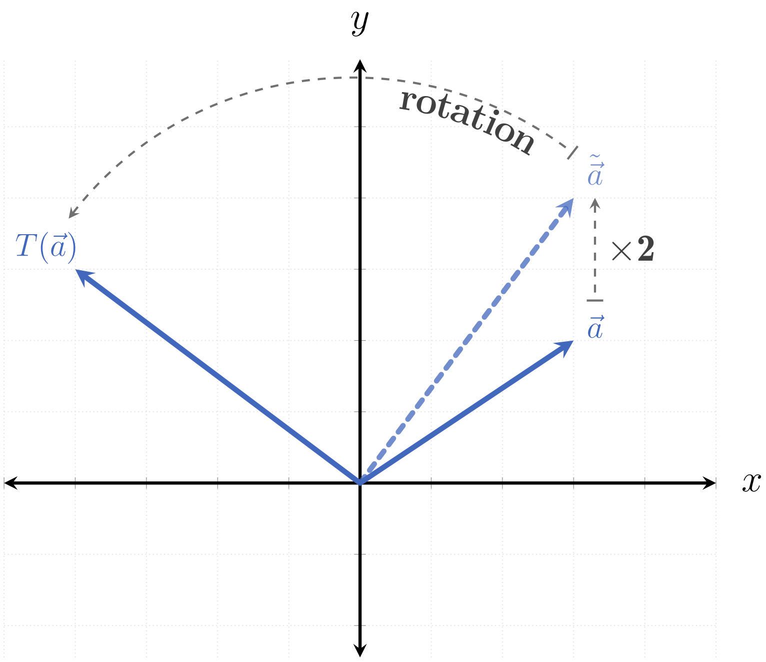


Figure 2.9: The transformation T applied to $\vec{a}$: first the vector is scaled by 2 in the y-direction, and then rotated by 90° anti-clockwise.

Now, what happens if we wish to calculate a more complicated linear transformation, which is both composed out of many more basic transformations - and also acts on higher-dimensional vectors? Performing each transformation step might not be so straight-forward, and is definitely not an efficient process. In fact, even if we're dealing with a simple example just like the one we just saw, we would still need to apply it every time we want to calculate how a new vector is transformed.

It turns out that thanks to the simplicity of linear transformations [2] we only need to perform a few simple calculations per transformation to practically get any possible application of it on any vector!

To see how this is the case, let's go back to our simple 2-dimensional example. First, recall that we can express any vector in terms of the standard basis vectors (REF), which in this case gives us (note the color-coding)

$$\vec{a}=\begin{bmatrix}3\\2\end{bmatrix}=3\overset{\substack{\hat{x}\\ \downarrow}}{\begin{bmatrix}1\\0\end{bmatrix}}+2\overset{\substack{\hat{y}\\ \downarrow}}{\begin{bmatrix}0\\1\end{bmatrix}}. \tag{2.36}$$

We can then apply the defining properties of linear transformations (REF) on $\vec{a}$:

$$\begin{aligned}
T\begin{bmatrix}3\\2\end{bmatrix}&=T\left(3\cdot\begin{bmatrix}1\\0\end{bmatrix}+2\cdot\begin{bmatrix}0\\1\end{bmatrix}\right)\\
&=T\left(3\cdot\begin{bmatrix}1\\0\end{bmatrix}\right)+T\left(2\cdot\begin{bmatrix}0\\1\end{bmatrix}\right)\\
&=3\cdot T\left(\begin{bmatrix}1\\0\end{bmatrix}\right)+2\cdot T\left(\begin{bmatrix}0\\1\end{bmatrix}\right)\\
&=3\cdot T\left(\hat{x}\right)+2\cdot T\left(\hat{y}\right).
\end{aligned} \tag{2.37}$$

Now, applying T to the two standard basis vectors $\hat{x}$ and $\hat{y}$ we get

$$\begin{array}{lllll}
\hat{x}=\begin{bmatrix}1\\0\end{bmatrix} & \xmapsto[\times 2\text{ in }y]{} & \begin{bmatrix}1\\0\end{bmatrix} & \xmapsto[90^\circ\text{ rotation}]{} & \begin{bmatrix}0\\1\end{bmatrix}=T(\hat{x})\\
\hat{y}=\begin{bmatrix}0\\1\end{bmatrix} & \xmapsto{\hspace{2em}} & \begin{bmatrix}0\\2\end{bmatrix} & \xmapsto{\hspace{3em}} & \begin{bmatrix}-2\\0\end{bmatrix}=T(\hat{y})
\end{array} \tag{2.38}$$

And when we use these transformed basis vectors in the linear combination we

[2] more correctly: to the strict rules they enforce

finally get

$$\begin{aligned} 3T\left(\hat{x}\right) + 2T\left(\hat{y}\right) &= 3\begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2\begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \end{aligned} \tag{2.39}$$

which is exactly what we got when we calculated T's effect directly on $\vec{a}$ (cf. Equation 2.41).

Notice exactly what happened: in the standard basis representation, $\vec{a}$ had the coefficients 3 and 2 for $\hat{x}$ and $\hat{y}$, respectively. When applying T to $\vec{a}$ these two coefficients remained, but the basis changed: instead of $\left\{\hat{x}, \hat{y}\right\}$, the two basis vectors became $\left\{T(\hat{x}), T(\hat{y})\right\}$.

Of course, this is not unique to $\vec{a}$. For example, let's look at another vector,

$$\vec{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

Its coefficients in the standard basis representation are 2 and -4, respectively. So we expect $T(\vec{b})$ to be

$$\begin{aligned} T(\vec{b}) &= 2T(\hat{x}) - 4T(\hat{y}) \\ &= 2\begin{bmatrix} 0 \\ 1 \end{bmatrix} - 4\begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 8 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 2 \end{bmatrix}. \end{aligned} \tag{2.40}$$

If we now calculate $T(\vec{b})$ directly, we get the following:

$$\vec{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \xmapsto{\times 2 \text{ in } y} \begin{bmatrix} 2 \\ -8 \end{bmatrix} \xmapsto{90^\circ \text{ rotation}} \begin{bmatrix} 8 \\ 2 \end{bmatrix} = T(\vec{b}). \tag{2.41}$$

What this insight shows us is that given a linear transformation, we only need to know how it transforms the respective standard basis vectors in order to calculate how it transform any vector in its domain. The transformed version of the vector would simply be the linear combination of these transformed basis vectors, with the coefficients given by the vector's presentation in the standard basis. It really can't get any simpler than that!

To make this a bit more formal, let's look at the most generic case: say $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation [3]. Then applying T to a vector $\vec{v} \in \mathbb{R}^n$ in its standard basis representation,

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

we get

$$\begin{aligned} T(\vec{v}) &= T\left(v_1 \hat{e}_1 + v_2 \hat{e}_2 + \cdots + v_n \hat{e}_n\right) \\ &= v_1 T(\hat{e}_1) + v_2 T(\hat{e}_2) + \cdots + v_n T(\hat{e}_n). \end{aligned} \tag{2.42}$$

This is an incredibly powerful insight which allows us to save a lot of calculations - and which we will use in the next section to make the calculations of linear transformations *even more* concise and simple, using the power of...

[3] we'll deal with the more general case of $T : \mathbb{R}^n \to \mathbb{R}^m$ where $n \neq m$ in the next section.

2.4 MATRICES

2.4.1 CALCULATING LINEAR TRANSFORMATIONS, PART 2

Following our example from the end of the previous section, let's look again at the result of applying T from Equation 2.41 on the vector $\vec{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$.

Since we know how T transforms the basis vectors $\hat{x}$ and $\hat{y}$ (Equation 2.38), we can $T(\vec{a})$ in the following way:

$$T(\vec{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \overset{\overset{a_x}{\downarrow}}{3} \underset{\underset{T(\hat{x})}{\uparrow}}{\begin{bmatrix} 0 \\ 1 \end{bmatrix}} + \overset{\overset{a_y}{\downarrow}}{2} \underset{\underset{T(\hat{y})}{\uparrow}}{\begin{bmatrix} -2 \\ 0 \end{bmatrix}} = \begin{bmatrix} 0 \cdot 3 - 2 \cdot 2 \\ 1 \cdot 3 + 0 \cdot 2 \end{bmatrix}. \tag{2.43}$$

Then, we collect all the coefficients in black into a single structure, which we call a matrix:

$$\mathbf{M} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}. \tag{2.44}$$

Finally, we define the action of the matrix $\mathbf{M}$ on the vector $\vec{a}$ as following:

$$\mathbf{M}\vec{a} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \cdot 3 - 2 \cdot 2 \\ 1 \cdot 3 + 0 \cdot 2 \end{bmatrix} = T(\vec{a}). \tag{2.45}$$

Of course, we should make sure that the matrix $\mathbf{M}$ we just defined works for other vectors as well. If we apply $\mathbf{M}$ to the second vector from our example, $\vec{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}$, we get

$$\mathbf{M}\vec{b} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ -4 \end{bmatrix} = \begin{bmatrix} 0 \cdot 2 - 2 \cdot -4 \\ 1 \cdot 2 + 0 \cdot -4 \end{bmatrix} = \begin{bmatrix} 8 \\ 2 \end{bmatrix} = T(\vec{b}), \tag{2.46}$$

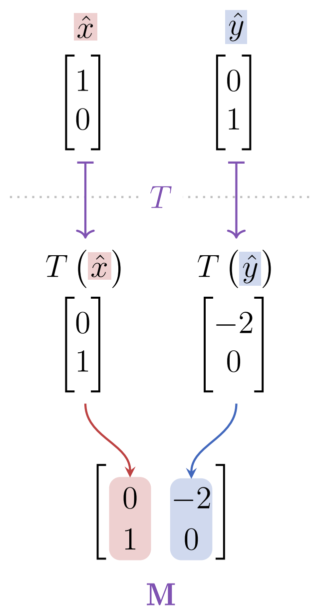
which is exactly the result we got in Equation 2.40. Great!

Why not continue checking how this matrix-vector product we developed works for other vectors? First, let's check $\hat{x}$ and $\hat{y}$ for which we already know the results (Equation 2.41):

$$\begin{aligned} \mathbf{M}\hat{x} &= \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 - 2 \cdot 0 \\ 1 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = T(\hat{x}), \\ \mathbf{M}\hat{y} &= \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \cdot 0 - 2 \cdot 1 \\ 1 \cdot 0 + 0 \cdot 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \end{bmatrix} = T(\hat{y}), \end{aligned} \tag{2.47}$$

exactly the results we expected.

Perhaps you've noticed something by now - the columns of $\mathbf{M}$ are exactly the vectors $T(\hat{x})$ and $T(\hat{y})$ in the right order:



The most general transformation we can have using $\mathbf{M}$ would be applying it to a general vector, $\vec{v} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$:

$$\mathbf{M}\vec{v} = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \cdot x - 2 \cdot y \\ 1 \cdot x + 0 \cdot y \end{bmatrix} = \begin{bmatrix} -2y \\ x \end{bmatrix}. \tag{2.48}$$

Of course, this result is exactly what we expect, as we can see when we apply each step of T on the generic vector $\vec{v}$:

$$\begin{bmatrix} x \\ y \end{bmatrix} \xmapsto{\times 2 \text{ in } y} \begin{bmatrix} x \\ 2y \end{bmatrix} \xmapsto{90^\circ \text{ rotation}} \begin{bmatrix} -2y \\ x \end{bmatrix}. \tag{2.49}$$

OK, up until now everything we did was for a specific transformation we called T. However, we can build a matrix for a general linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$. Such a transformation has the form

$$T\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix},$$

where $a, b, c, d \in \mathbb{R}$ (REF), and using the same process as in Equation 2.43 we get the general matrix

$$\mathbf{M} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}. \tag{2.50}$$

Just like in our specific example, the columns of $\mathbf{M}$ are exactly the images of the basis vectors $\hat{x}$ and $\hat{y}$, respectively. To confirm this, let's apply the $\mathbf{M}$ on both basis vectors:

$$\begin{aligned} \mathbf{M}\hat{x} &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a \cdot 1 + b \cdot 0 \\ c \cdot 1 + d \cdot 0 \end{bmatrix} = \begin{bmatrix} a \\ c \end{bmatrix} = M_1, \\ \mathbf{M}\hat{y} &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} a \cdot 0 + b \cdot 1 \\ c \cdot 0 + d \cdot 1 \end{bmatrix} = \begin{bmatrix} b \\ d \end{bmatrix} = \mathbf{M}_2. \end{aligned} \tag{2.51}$$

Note that I use M_1 and M_2 to denote the 1-st and 2-nd columns of $\mathbf{M}$, respectively.

In most textbooks and courses I've seen, the process of multiplying a matrix $\mathbf{M}$ by a vector $\vec{v}$ is described as following: the i-th component of the resulting vector (in our case $i \in \{1, 2\}$) is the *dot product* of the i-th row of $\mathbf{M}$ and the vector $\vec{v}$ itself. This is exactly what happens in practice in the development process we went through here - if we name the resulting vector of $\mathbf{M}\vec{v}$ as $\vec{w}$ we get its following two components:

$$\left.\begin{aligned} w_1 &= \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} a \cdot x + b \cdot y \end{bmatrix} = \begin{bmatrix} ax + by \end{bmatrix} \\ w_2 &= \begin{bmatrix} c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c \cdot x + d \cdot y \end{bmatrix} = \begin{bmatrix} cx + dy \end{bmatrix} \end{aligned}\right\} \to \vec{w} = \begin{bmatrix} ax + by \\ cx + dy \end{bmatrix}. \tag{2.52}$$

Of course, this is just the case for a matrix representing a general linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$. Matrices representing such transformations have 4 components, since they are made of 2 basis vectors as columns, each having 2 components (since the vectors live in $\mathbb{R}^2$). We also say that the matrix lives in the space $\mathbb{R}^{2\times 2}$.

In general (also in higher-dimensional cases), we use 2 indices to denote the components of any matrix: the 1st index for the *row* of the component and the 2nd index for the *column* of the component. For example, a general matrix $\mathbf{A} \in \mathbb{R}^{2\times 2}$ can be written as

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \tag{2.53}$$

where it is important to remember that there are two indices here - and not a single number. For example, a_{11} should be read as "a-one-one" (i.e. the component in the 1st row and the 1st column), and not as "a-eleven". Similarly, a_{12} is "a-one-two" (not "a-tweleve"), a_{21} is "a-two-one", etc. The reason this notation is used (as much as it might be confusing at first) is lazyness: nobody wants to write redundant commas (i.e. $a_{1,1}$). It is what it is, and you will get used to it ¯_(ツ)_/¯

We can now finally move on to the most general form of a linear transformation between a real vector space to itself, i.e. $T : \mathbb{R}^n \to \mathbb{R}^n$. The most general vector $\vec{v} \in \mathbb{R}^n$ has the form

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}, \tag{2.54}$$

and therefore the application of a generic linear transformation from $\mathbb{R}^n$ to itself on $\vec{v}$ yields the vector

$$T(\vec{v}) = \vec{w} = \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{n1}v_1 + a_{n2}v_2 + \cdots + a_{nn}v_n \end{bmatrix}, \tag{2.55}$$

where $a_{ij} \in \mathbb{R}$ (with $i, j \in \{1, 2, \ldots, n\}$). Of course, denoting the components as a_{ij} is alluding to the next step - collecting them into a matrix:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}. \tag{2.56}$$

Just like in the 2×2 case, the i-th column vector of $\mathbf{A}$ shows how it transforms $\hat{e}_i$ (which in this case lives in the space $\mathbb{R}^n$, of course):

$$\mathbf{M} = \begin{array}{cccc} T(\hat{e}_1) & T(\hat{e}_2) & & T(\hat{e}_n) \\ \downarrow & \downarrow & & \downarrow \end{array} \atop \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix}.$$

Therefore, in general we define the product of an $n \times n$ matrix $\mathbf{A}$ (otherwise written as $\mathbf{A} \in \mathbb{R}^{n \times n}$) and a vector $\vec{v} \in \mathbb{R}^n$ to be

$$\mathbf{A}\vec{v} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n \\ \vdots \\ a_{n1}v_1 + a_{n2}v_2 + \cdots + a_{nn}v_n \end{bmatrix} = \vec{w}. \tag{2.57}$$

Indeed, this can be presented (as it often does) in the following way: the i-th coefficient of the vector $\vec{w} = \mathbf{A}\vec{v}$ is the dot product of the i-th row of $\mathbf{A}$ and the vector $\vec{v}$. To see this in practice, follow closely how each of the rows $1, 2$ and n of $\mathbf{A}$ in Equation 2.57 are dot-multiplied with $\vec{v}$ to yields the corresponding component of $\vec{w}$.

Note 2.1 Square Matrices

All the kinds of matrices we defined so far are called square matrices, since they have an equal number of rows and columns (since they map from any vector space $\mathbb{R}^n$ to itself) - i.e. their dimensions are of the form $n \times n$.

However, in general matrices can represent any kind of linear transformation - also those that map between different real vector spaces, i.e. $\mathbb{R}^n \to \mathbb{R}^m$ (where $n \neq m$). These matrices would be of order $m \times n$, and are of course **not** square matrices. However, since square matrices are the easiest to understand when first learning the topic, and they also happen to play a very important role in linear algebra - I chose to concentrate on them in the beginning, before moving on to more general matrices.

!

2.4.2 THE FUNDAMENTAL LINEAR TRANSFORMATIONS AS MATRICES

Let us now use the process we devolped above to represent the fundamental $\mathbb{R}^2 \to \mathbb{R}^2$ linear transformations (REF) in matrix form. Since we're dealing with transformations from $\mathbb{R}^2$ to itself, the matrices representing these transformations are of dimension 2×2, i.e. they live in the space $\mathbb{R}^{2\times 2}$.

We start with the most boring linear transformation: the identity transformation I. As a reminder, I maps *any* vector $\vec{v} \in \mathbb{R}^2$ to itself: $I(\vec{v}) = \vec{v}$. In particular, it takes the two standard basis vectors $\hat{x}$ and $\hat{y}$ and maps them to themselves:

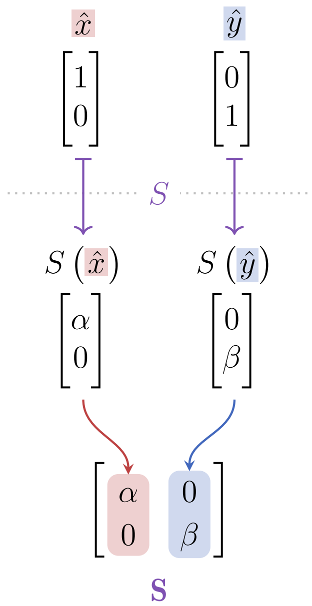
$$\begin{bmatrix}1\\0\end{bmatrix} \overset{I}{\mapsto} \begin{bmatrix}1\\0\end{bmatrix}, \quad \begin{bmatrix}0\\1\end{bmatrix} \overset{I}{\mapsto} \begin{bmatrix}0\\1\end{bmatrix}. \tag{2.58}$$

Very exciting indeed. Knowing how I transforms $\hat{x}$ and $\hat{y}$ we can use their maps as the columns of the matrix representation of I:

$$\begin{array}{ccc} \hat{x} & & \hat{y} \\ \begin{bmatrix}1\\0\end{bmatrix} & & \begin{bmatrix}0\\1\end{bmatrix} \\ \Big\downarrow & I & \Big\downarrow \\ I\left(\hat{x}\right) & & I\left(\hat{y}\right) \\ \begin{bmatrix}1\\0\end{bmatrix} & & \begin{bmatrix}0\\1\end{bmatrix} \end{array} \quad \longrightarrow \quad \underset{\mathbb{I}_2}{\begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}} \tag{2.59}$$

($\mathbb{I}$ is a common notation for identity matrices, and the index 2 signifies that this is the specific 2×2 identity matrix)

The next type of fundamental linear transformations we should represent in matrix form are the scaling matrices. Instead of calculting their representation separately, let's directly find the matrix representation of the linear transformation S which scales any vector by α in the x-direction and by β in the y-direction:



(2.60)

$\mathbf{S}$ has an important property: it only has non-zero entries in a_{11} and a_{22}: we call these elements together the main diagonal of the matrix. A matrix which has non-zero entries only in its main diagonal is unsurprisingly called a diagonal matrix[4].

Note 2.2 The Identity Matrix is a Diagonal Matrix

In the specific case where $\alpha = \beta = 1$ a scaling matrix essentially does nothing: it scales by 1 in both directions. Indeed, in this case it becomes the identity matrix $\mathbb{I}_2$.

!

The next family of linear transformations to represent as matrices are the rotations around the origin. Again, we don't really need to think too much: just check the results of rotating $\hat{x}$ and $\hat{y}$ by θ anti-clockwise. Since $\hat{x}$ is a unit vector, rotating it by θ anti-clockwise around the origin gives another unit vector (since rotating a vector doesn't change its norm), which is at an *theta* to the x-axis. Its coordinates are then exactly the polar coordinates of a vector at θ to the x-axis but with $R = 1$

[4] this might seem unremarkable now, but in higher dimensions these matrices are very important.

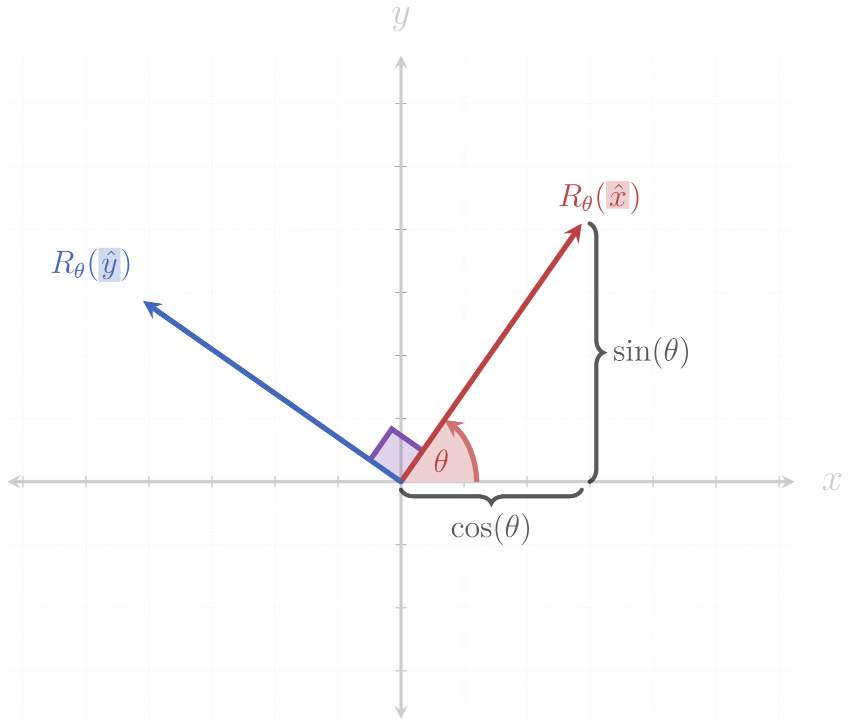


Figure 2.10: The two basis vectors $\hat{x}$, $\hat{y}$ rotated by an angle θ anti-clockwise around the origin. The coordinates of the rotated $\hat{x}$ are simply the polar coordinates of a general vector which is at an angle θ to the x-axis, with $R = 1$ (since $\hat{x}$ is a unit vector).

(see REF TO POLAR COORDINATES EQUATION):

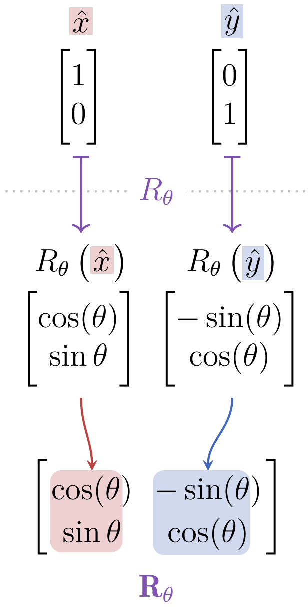
$$R_\theta(\hat{x}) = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}. \tag{2.61}$$

Since the rotated $\hat{y}$ is orthogonal to the rotated $\hat{x}$ we can use the orthogonality trick in 2-dimensions to find its coordinates (REF TO ORTHOGONALITY TRICK IN 2D):

$$R_\theta(\hat{y}) = \begin{bmatrix} -\sin(\theta) \\ \cos(\theta) \end{bmatrix}. \tag{2.62}$$

(see Figure 2.10 for a visualization of this idea)

Therefore, the rotation matrix $\mathbf{R}_\theta$ is



(2.63)

Let's check some specific examples of rotationg matrices, starting with $\theta = 0$. This gives the matrix

$$\mathbf{R}_0 = \begin{bmatrix}\cos(0) & -\sin(0)\\ \sin(0) & \cos(0)\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = \mathbb{I}_2, \tag{2.64}$$

which is exactly what we expect, since a rotation by $\theta = 0$ doesn't do anything, so we get back the identity matrix.

A rotation by $\theta = \frac{\pi}{2}$ (equivalent to $90°$) is done by the matrix

$$\mathbf{R}_{\frac{\pi}{2}} = \begin{bmatrix}\cos\left(\frac{\pi}{2}\right) & -\sin\left(\frac{\pi}{2}\right)\\ \sin\left(\frac{\pi}{2}\right) & \cos\left(\frac{\pi}{2}\right)\end{bmatrix} = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}. \tag{2.65}$$

This is an nice result: rotating by $\frac{\pi}{2}$ is equivalent to switching the two basis vector $\hat{x}$ and $\hat{y}$ and then flipping the "new" $\hat{y}$ around so it faces in the negative x-direction.

Before continuing to $matR_\pi$ let's see what we expect: rotating a vector by $\theta = \pi$ is simply inverting it (i.e. negating both its coordinates). And indeed,

$$\mathbf{R}_\pi = \begin{bmatrix}\cos(\pi) & -\sin(\pi)\\ \sin(\pi) & \cos(\pi)\end{bmatrix} = \begin{bmatrix}-1 & 0\\ 0 & -1\end{bmatrix}, \tag{2.66}$$

which is exactly the identity matrix with all its entries negating. How cool is that?

Ok, next up: reflections! Reflecting a vector across the x-axis is simply flipping it[5] vertically - so its x-component doesn't change, and its y-component gets flipped. The result for the basis vectors is that $\hat{x}$ stays the same (i.e. reflecting $\hat{x}$ across the x access doesn't change it), while $\hat{y}$ gets flipped:

$$
\begin{tikzcd}
\overset{\hat{x}}{\begin{bmatrix}1\\0\end{bmatrix}} \arrow[d, mapsto] \arrow[r, phantom, "\mathrm{Ref}_x"] & \overset{\hat{y}}{\begin{bmatrix}0\\1\end{bmatrix}} \arrow[d, mapsto] \\
\overset{\mathrm{Ref}_x\left(\hat{x}\right)}{\begin{bmatrix}1\\0\end{bmatrix}} & \overset{\mathrm{Ref}_x\left(\hat{y}\right)}{\begin{bmatrix}0\\-1\end{bmatrix}}
\end{tikzcd}
\qquad
\underbrace{\begin{bmatrix}1 & 0\\0 & -1\end{bmatrix}}_{\mathrm{Ref}_x}
\tag{2.67}
$$

Similarly, reflecting a vector across the y-axis flips their x-coordinate and keeps

[5]yes, I also laughed when writing this.

their y-coordinate untouched, i.e. $\hat{x}$ gets flipped and $\hat{y}$ remains the same:

$$
\begin{tikzcd}
\hat{x} & & \hat{y} \\
\begin{bmatrix}1\\0\end{bmatrix} \arrow[d] & \mathrm{ref}_y & \begin{bmatrix}0\\1\end{bmatrix} \arrow[d] \\
\mathrm{ref}_y\left(\hat{x}\right) & & \mathrm{ref}_y\left(\hat{y}\right) \\
\begin{bmatrix}-1\\0\end{bmatrix} \arrow[dr, bend right] & & \begin{bmatrix}0\\1\end{bmatrix} \arrow[dl, bend left] \\
& \underbrace{\begin{bmatrix}-1 & 0\\0 & 1\end{bmatrix}}_{\mathbf{ref}_y} &
\end{tikzcd}
\tag{2.68}
$$

See **??** for a visualization.

Before we can build a general reflection matrix (i.e. representing a reflection across *any* line going through the origin, not only the axes) we need to learn a couple of more things, so we'll skip that for now and hop on to building matrices for shear transformations.

Now for the shear transformations: a shear in the x-direction keeps $\hat{x}$ untouched, while simply moving $\hat{y}$ by some amount in the x-direction (see Figure 2.11).

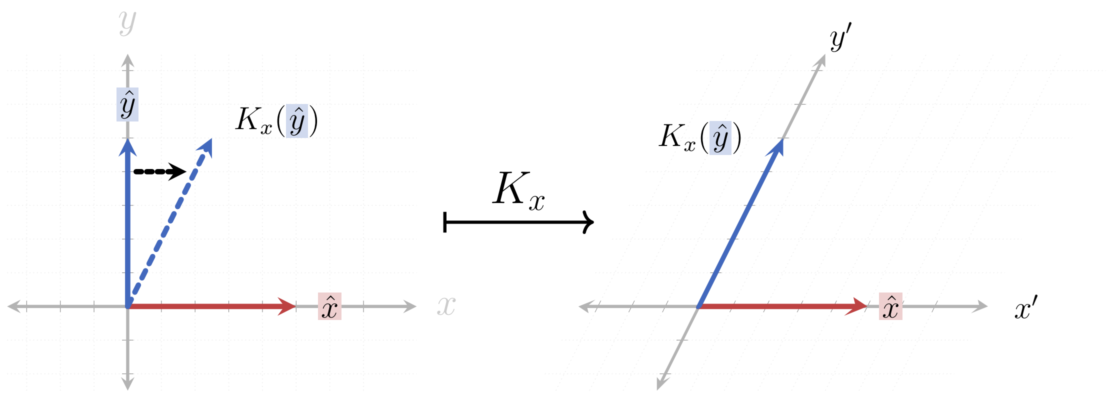


Figure 2.11: Shearing in the x-direction: while $\hat{x}$ is kept untouched, $\hat{y}$ is shifted by some value $k \in \mathbb{R}$.

2.5 LINEAR EQUATIONS

2.6 EIGENVECTORS AND EIGENVALUES

2.7 MATRIX DECOMPOSITIONS

$$T(\alpha\vec{x} + \beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} \mid \vec{b} \rangle = \vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i$$

$$\vec{a} \cdot \vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

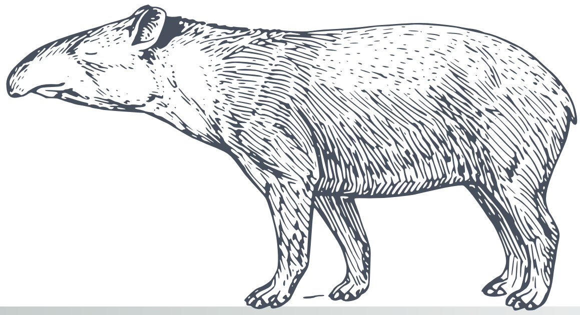
$$(\mathbf{A} \cdot \mathbf{B})^\intercal = \mathbf{B}^\intercal \cdot \mathbf{A}^\intercal$$

$$\vec{v} = \alpha_1 \hat{e}_1 + \alpha_2 \hat{e}_2 + \dots + \alpha_n \hat{e}_n$$

$$\mathbf{A}\vec{x} = \lambda\vec{x}$$

$$\det(\mathbf{A} \cdot \mathbf{B}) = \det(\mathbf{A}) \cdot \det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A} \cdot \mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

3

INTRODUCING SOME FORMALISM

$$T(\alpha\vec{x}+\beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle \vec{a} \mid \vec{b} \rangle = \vec{a}\cdot\vec{b} = \sum_{i=1}^{n} a_i b_i$$

$$\vec{a}\cdot\vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

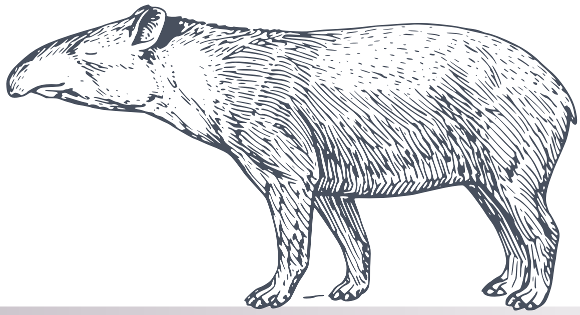
$$(\mathbf{A}\cdot\mathbf{B})^{\intercal} = \mathbf{B}^{\intercal}\cdot\mathbf{A}^{\intercal}$$

$$\vec{v} = \alpha_1\hat{e}_1 + \alpha_2\hat{e}_2 + \dots + \alpha_n\hat{e}_n$$

$$\mathbf{A}\vec{x} = \lambda\vec{x}$$

$$\det(\mathbf{A}\cdot\mathbf{B}) = \det(\mathbf{A})\cdot\det(\mathbf{B})$$

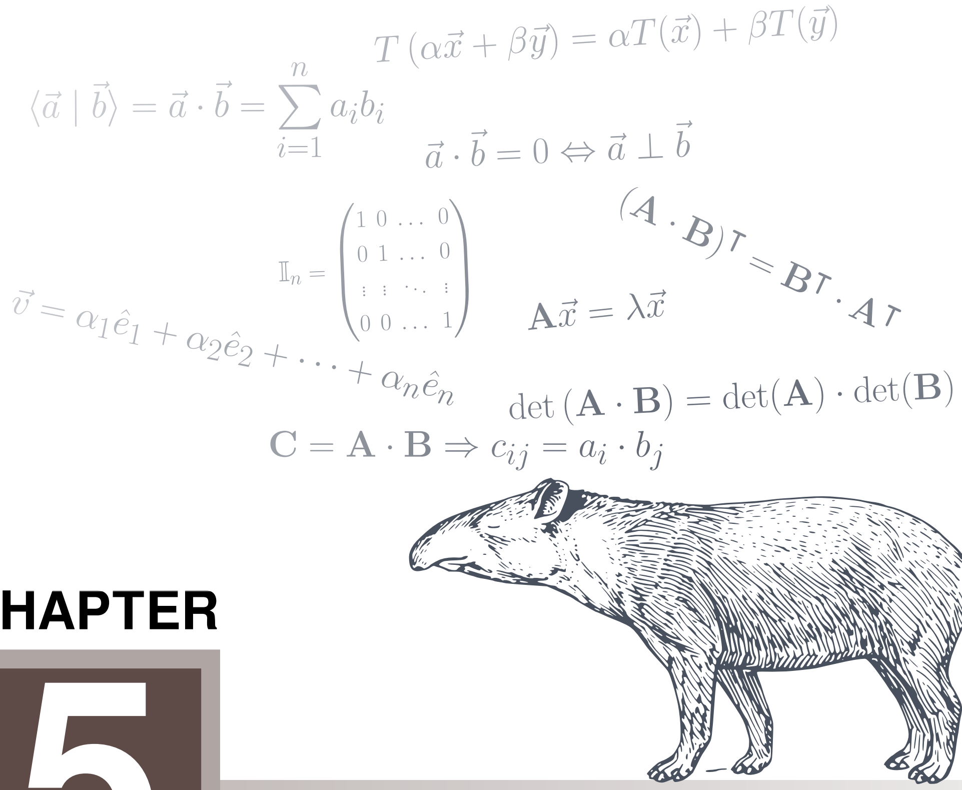
$$\mathbf{C} = \mathbf{A}\cdot\mathbf{B} \Rightarrow c_{ij} = a_i \cdot b_j$$



CHAPTER

4

ADVANCED TOPICS IN LINEAR ALGEBRA



CHAPTER

5

FURTHER RELATED TOPICS

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